

DEPARTMENT OF COMPUTER SCIENCE

Exploring the Potential of AI Integration into the Comic Creation Process

Pragya Gurung

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Abstract

The use of artificial intelligence (AI) has been steadily increasing across many industries, including the comic book industry. While there have been numerous studies exploring people's perceptions of AI and AI-based creativity support tools(AI-CSTs), there has been a lack of studies on how AI can be integrated into the comic creation process and how comic creators interact with such tools. This is unfortunate because comic books are a unique graphic art form that combines both illustration and text, making them an ideal medium to leverage AI to potentially enhance the creative process. To address this, in this project I developed ComicPal, an AI-assisted creativity support tool designed to facilitate the creation of comic books, then recruited creators to use the tool and discuss their experiences. The system was evaluated with questionnaires and a focus group to determine its usability and effectiveness. Thematic analysis was performed on the qualitative data collected during the studies to identify patterns in the interaction dynamics between creators and AI. From my findings, I found that AI can be a valuable tool for ideation and inspiration, especially when facing creativity blocks or gaps. However, creators' attitudes towards AI assistance varied depending on their current values and motivations, which were influenced by factors such as the task they were performing or their personal feelings towards their project and skills. Additionally, although most creators appreciated the benefits of AI, they all had reservations rooted in personal or anecdotal negative experiences. The results of this project was then used to discuss potential design considerations for future development of AI-CSTs for comic creation. Along with suggestions for further work to explore potential research areas and possible enhancements for ComicPal to support collaborative creation.

Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Pragya Gurung, Saturday 22nd July, 2023

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Ethics Statement

This project fits within the scope of ethics application 0026, as reviewed by my supervisor, Dr Peter Bennett and Sarah Connolly

Chapter 1

Introduction

There has been a growing familiarity with the concept of Artificial Intelligence (AI) among the general public [75, 54, 57]. A universally accepted definition of AI is not clear but it is typically described as "The performance of tasks, which, if performed by a human, would be deemed to require intelligence" [81]. AI systems such as DALL-E [18], Midjourney [55] and ChatGPT [11] have become popular and a common talking point amongst those in many industries, including the comic book industry, where some creators have already started to use and experiment with AI in their creative process [63, 34, 31]. As a result there has been increasing discussion about the ethics and copyright concerns of using AI as a creativity tool [30]. For instance, a recent case in 2023 involved a creator losing all US Copyright protection for the Midjourney generated images used in their comic book [28]. Unfortunately, the exploration of AI in the comics industry has largely being led by individual or communities of dedicated creators, rather than holistic research studies. While AI-powered creativity support tools (AI-CSTs) have been investigated in how they assist creative writing [33] and collaborative drawing [59], the realm of AI-CSTs for comic creation has largely been under explored, with only a few studies conducted such as [82]. Given that comic books are a graphic art form which portrays a story through a deliberate sequence of illustrations and text [29], there exists significant potential for AI integration into the creative process, making it an intriguing research field that remains largely untapped.

My project aims to explore the ways in which AI-assistance can enhance comic creators' workflows and how creators interact with the AI. To achieve this I wanted to collect first hand data on comic creators who have experienced using AI in their workflow; therefore, I developed a bespoke tool called ComicPal (Appendix A) for the participants of my studies to use. This project has two main phases - ComicPal Development and Focus Group and Research Findings. During the first phase, I developed a webapp called ComicPal using an iterative design process where I frequently sought feedback and conducted user testing to improve the tool. The webapp makes use of OpenAI's API to help users create stories, characters and comic pages. Fifteen participants were recruited for the user testing. They used the webapp to create a short 1-2 page comic and then completed a questionnaire. The results from this questionnaire were then used to inform the further development of ComicPal.

The second phase of the project involves conducting a focus group study with four creators. These creators were once again tasked to individually create a short 1-2 page comic, and then discuss their experiences in using AI in their creative process. A transcript was made and thematic analysis was performed to extract overarching themes. The themes indicated that creators wants and needs can influence their interaction dynamics with the AI. However, what these needs were dependent on many factors such as the current task being performed or the content of the project. Additionally, though creators appreciated the AI's ability to help inspire new ideas, they all held reservations about using the technology as a creativity tool that were fuelled by personal or anecdotal negative experiences.

Finally, with the findings of this project, I propose some design considerations for future AI-CST tools catered towards comic book creation.

1.1 Aims and objectives

The goal of this research project is to explore how AI can be incorporated into the workflows of comic book creators, thereby developing an understanding of what factors facilitate better or worse human-AI collaboration in the context of comic book creation. The aims and objectives were to:

- Conduct a literature review on the use of AI in creative industries, with a focus on the comic industry.
- Identify the challenges and limitations of being a creator in the comic industry and investigate solutions using AI to address them.
- Develop an AI-assisted creativity support tool called ComicPal.
- Perform user testing to gather data and to ensure the system is an effective tool for comic creators.
- Analyse the data collected from testers to assess the impact of AI on creativity in the comic creation process.
- Develop possible design considerations for future AI-CSTs geared towards comic creators.

Chapter 2

Background

This chapter will provide an overview of the related work and literature that is relevant to this project. It will begin by defining key concepts, followed by an exploration of the various factors that affect people's perceptions of AI. The chapter will then delve into the topic of Human-AI co-creativity and concludes with a discussion about the current state of AI support for comic creation. By conducting this review, I aim to highlight the key gaps and opportunities for exploration within the existing literature, thereby establishing a foundation for my project that follows.

2.1 Defining AI

What AI is defined as has been debated by psychologists and scientists, however it has been a challenge to arrive at a consensus [46]. The most widely recognised definition is attributed to Marvin Minksy [81] and it is the definition that I will adopt for this project. "The performance of tasks, which, if performed by a human, would be deemed to require intelligence."

Tasks that require human intelligence include memory, attention, processing speed, and executive functions (i.e., reasoning, planning, problem solving, and multitasking) [4]. To illustrate the differences between an AI and non-AI system, I will make the comparison between Strandbeests, which are kinetic sculptures made by Dutch artist Theo Jansen, and a bipedal robot developed in [41]. Jansen's sculptures, although incredible, only possess the mechanical function to walk. On the other hand, the bipedal robot assesses the world around it and adapts to the environment when walking. It essentially learns by solving the problem of achieving "reliable locomotion in demanding environments with varied terrain and geometric constraints" [41].

For the scope of this research, I will only be considering the current state of AI technology as of the time of writing this project. Therefore, I will not take into account any technologies that may still be in the development stage.

2.2 Comic Books

For the purpose of this project I will define comic books as a graphic art form which portray stories through a deliberate sequence of illustrations and text containing dialogue, sounds or narratives [29]. A particular story may be told in a single issue or a series of issues and typically have one or more central characters who appear in each issue. Comic production has gradually transitioned from paper-based workflows towards fully digital ones [52], as readers are increasingly using the web and mobile platforms to consume them. Typically, a team of individuals work together to produce a single comic, with specialised roles such as writer, penciller, inker, colorist, letterer and editor [44]. The stages [44] from conception to creation include:

1. Ideation/Concept
2. Plot Development
3. Script
4. Art Production

5. Pencils
6. Inks
7. Colours
8. Lettering
9. Editorial
10. Printing
11. Marketing
12. Distribution

This is a very broad generalisation, and these roles are usually carried out by teams of different individuals, although the creative aspect can also be carried out by a single creator. Combining the roles of plot development, writing and drawing is easier than combining writing and distribution, particularly following the transition of distribution primarily being through newsstands and local shops to a centralised distribution model through specialist comic shops [52]. In addition, many of the stages listed above will have different phases in of itself. For example, the plot development stage would also include a phase where the characters and their backstories are developed.

Moreover, changing technologies and the rise in digitised comics [52] have enabled a new form of comics - the webcomic - which enables more solo creators to handle most, if not all of these jobs. For example an artist would handle both pencilling and inking duties, and possibly even colouring. Once edited, the comic is either submitted to a third party (for example, LINE Webtoon or Tapas) to be distributed or is hosted by the creators themselves on personal website [80].

2.3 Perception of AI

As this project aims to examine how creators can use AI to aid in the comic creation process, I found literature that explores what factors seem to influence people's perception of AI and therefore their usage of it. Such factors include psychological, social and cultural.

Some research indicates that individual's personality traits and demographics affect their attitude towards AI [72] with suggestions that fear and distrust of AI is associated with neuroticism. In contrast, openness and confidence was related to AI acceptance. Such sentiments were also found to be influenced by demographic factors as older adults tended to view AI as less useful and dangerous than younger demographics [5]. In addition, existing familiarity or experience with AI technologies was another factor impacting users perception. For example, medical students who were aware of AI being discussed in radiology and were more "tech-savvy", were more likely to agree that AI could potentially make an automated diagnosis from imaging examinations [65]. Thereby suggesting that as familiarity with AI technology increases, so does the confidence people have in AI.

Additionally, the way AI is portrayed in pop culture such as movies, books and games has also shown to impact people's perception of it, and these medias have influenced viewers understanding of AI's capabilities and potential impact [58]. Interestingly, those who believe AI as being realistically depicted in the media were more likely to think of AI as having the potential to feel emotion than to imagine AI replacing humans in jobs or operating as surveillance tools [58].

2.4 Human-AI Co-creativity

Creativity has been a part of many academic, professional and artistic domains and is a topic that has been studied both extensively and heterogeneously [6]. Some common characteristics across disciplines have arisen such as 'Agency', 'Support and Scaffolds' and 'Artefacts' [6]. 'Agency' refers to who or what causes creative activity such as an individual or a social group. 'Support and Scaffolds' refer to external characteristics that appear to support creativity, such as tools and methods. 'Artefacts' alludes to how a creative outcome can be identified and evaluated.

2.4.1 Creativity Support Tools

Creativity Support Tools(CSTs) is a field of research with the goal of developing improved software and tools to empower users to be more productive and innovative [49, 71] There is a substantial body of literature which explores both digital and analogue CSTs [13, 71, 17]. For example, [40] explores how to better support experiences of failures for novice creators and how creativity tools can provide a space to experiment and fail. However with the emergence of AI, there has been increasing interest in the possibilities of AI-powered CSTs(AI-CSTs). For example, [12] explores different types of AI-CSTs and defines guidelines for designing user expression interactions for the tools.

2.4.2 Limitations

There are various limitations to working with AI technologies, such as the lack of transparency and reproduction. This constitutes both a practical and ethical issue as explanations are often demanded by users who may not have a deep technical understanding of AI. Explainable AI (XAI) is a field of research that attempts to resolve this [47]. Good collaborative creativity requires a sustained back and forth dialogue and interactions between collaborators [48]. However, unlike with human to human collaboration, human and computer collaboration can often feel restrictive and shallow. As a result, it could lead to situations where the human is struggling to keep the AI "alive" [33] in an attempt to revitalise a productive dialogue. There is a lack of support for negotiations and discussion when working with computational creativity as interactions are carried without any (or little) user intervention [48]. XAI's could bridge this gap to foster fruitful co-creation with sustained continuous dialogues and interaction. As AI technologies advance over time, such collaborative co-creation between humans and AI also improves thereby broadening the AI's role as a collaborator in many creative areas such as storytelling [9, 20].

Natural language models such as GPT-3 that facilitate continuous dialogues between humans and AI are becoming increasingly common. For example, there is Googles LaMDA - Language Model for Dialogue Applications [43] and BLOOM - BigScience Language Open-science Open-access Multilingual [8]. Additionally, more sophisticated image generation models have begun to see wide spread use [55, 18] and the outputs of these models have been getting better exponentially [56]. To the point where AI generated artworks are winning against human made art in competitions [68].

2.5 AI Support for Creative Writing, Art and Comics

In this project, I will be developing ComicPal, an AI-assisted tool for comic creation. To inform the development of this tool, I conducted extensive research into the current landscape of AI support that exists for comic creation. While I found one existing tool, Neural Canvas, designed specifically for comic creation, I discovered that there is a lack of AI creativity support tools in this domain. Therefore, I decided to focus my research on exploring how AI can aid the various stages of comic creation as found in 2.2.

In this section I will highlight the relevant literature that explores the use of AI to help with drawing artwork, colourisation and creative writing. Additionally, I will discuss other notable programs or software that are relevant to this project.

2.5.1 Drawing

There are a variety of CSTs focused on supporting artistic creativity with the aim of coaching the user's technical skills. For example, ShadowDraw [45] was designed to assist users in drawing an object by providing them advice in real-time to help them draw more accurately. Similarly, the tool iCanDraw [24] provides feedback to users when drawing portraits to enhance their accuracy. However, tools such as these do not meet the needs of users who are looking to derive enjoyment from the creative process itself. Therefore, a new set of CSTs has emerged called "Casual Creators" that are designed with the aim to provide an enjoyable creative experience and instead of just focusing on creating a high-quality end-product [16]. An example of such a tool is Drawing Apprentice [21], a co-creative drawing partner. Another example is DuetDraw developed by Changhoon Oh et al. [59], an AI interface that allows users and an AI agent to draw collaboratively in real time. Changhoon Oh et al. conducted a study where they found that users consistently wanted to take the lead when drawing and wanted the AI to explain its intentions, but only when the user asked for it.

2.5.2 Colourisation

In some cases where human collaborators relationships are much simpler, it may be simpler to add CST [14]. For example, in the beginning of the colourisation stage of the comic book creation process, the colourist has to lay down the base colours for the drawings, this is called 'flatting' [35]. It can be a labour intensive and menial task, therefore, comic artists are more likely to readily accept a tool to automate it [14]. One such tool is FlatMagic [82], developed by Chuan Yan et al. It's an interactive, AI-driven flat colourisation support tool designed for use in Photoshop. From their findings, professionals who used the tool found it "useful and usable" [82] thereby suggesting a focus on partial automation may lead to better, more accepted, AI-CSTs .

2.5.3 Writing

There has been significant interest in designing computer assisted writing tools, however the focus of most of these tools revolves around idea generation [67, 42, 15]. Some examples of tools include Creative Help [67] which generates suggestions for the next sentence of a story as it's being written by the user, and Multiverse [42] where users choose from GPT-3 AI-generated suggestions iteratively until a tree-like structured narrative was formed. In addition, a tool developed by John Joon Young Chung et al. called TaleBrush [15], provided users a unique method of generating ideas because it used drawn sketches of a character's fortune as input. The tool then generates corresponding story sentences. Their research found that participants could understand the AI-generated output and use it to explore and write stories according to their own creative intentions, whilst also drawing inspiration from the generated ideas.

2.5.4 Other Tools

As mentioned at the start of this subsection, Neural Canvas was the only AI-tool I found that was designed specifically for comic creation. However, rather than a creativity tool, I found it to be a comic book generator where users just needed to pass in a few parameters about the plot, art style and overall mood and the tool will create the entire comic for them. The creators input starts and ends in the beginning of the process and the rest is left to the tool to finish. There is no sense of collaboration or negotiation as the creator can only determine if the comic is satisfactory once it's been fully made by the AI.

Other AI-tools that have seen increasing use for creating comics include image generators such as OpenAI's DALL-E [18] and Midjourney [55]. While not specifically for comic creation, comic creators have been experimenting with these tools to generate artwork for characters, covers and panels [31, 3, 34, 63]. In contrast to Neural Canvas, using image generators gave creators more creative control. Typically, creators would generate numerous images for a single panel and then meticulously sift through them to find the best one [34]. The AI-generated image selected is often the one used for the comic, with only minor adjustments made using Photoshop.

2.6 Summary

From the literature I have examined so far, I have found that although research on the integration of AI in writing and art has been conducted, the application of AI in the realm of comic books has been largely unexplored. While creating a comic book shares similarities with both creative writing and art, the medium presents its own unique set of challenges that could be worth exploring with AI as a creative support tool.

Chapter 3

Defining The Problem

3.1 Challenges Faced By Comic Creators

In the realm of creative work, including comic creation, encountering creative blocks is a common occurrence [1]. Creative blocks refer to the point at which one becomes stuck or deprived of ideas during the product development process [27]. Creativity is described by authors Jamie Ward et al. in [79] as "the ability to generate novel associations that are adaptive in some way". It is often driven by the combination, transformation, and exploration of existing ideas, but can become inaccessible due to a lack of new ideas [27], thereby, stifling individuals creative flow [26]. The White Page Syndrome is one type of creative block commonly faced by comic creators [1], characterised by the fear and anxiety associated with starting a new project. Questions such as "What if my work turns out badly?", "What if I have to start over?" would plague creators mind, paralysing them from initiating the development process. In [1] and [60], the authors propose a number of strategies for overcoming creative block. For instance, to alleviate the burden of developing one's own ideas, writers could follow other people's writing prompts or perform free-writing, which involves writing down as many ideas as possible uninterrupted for 10-15 minutes. For visual artists, the authors suggested they engage with less committed and easily restartable creative activities. For example, artists could use a digital canvas instead of a physical one, as it allows for unlimited undo and redo options, reducing the fear of ruining one's work and losing money from having to start over.

Another challenge that comic creators commonly encounter is burnout [1]. Originally coined by [32], burnout refers to the psychological response to prolonged exposure to high levels of stress. Unlike creative blocks where creators want to create but are unable to do so, burnout results in a lack of desire or the inability to create. Some solutions to treat/prevent burnout is suggested in [1] and [51], such as changing work patterns by taking more breaks or engaging in creative activities in different ways.

Comic creators also face a unique challenge of balancing words and images in comics [2]. To achieve this, it requires a deep understanding of both writing and visual storytelling, as well as the ability to seamlessly integrate the two aspects in an organic way. Additionally, how the text and images are laid out on the page is also crucial. Comic creators must master the use of space, or lack thereof, to convey ideas like the passage of time or suspense [36, 53].

3.2 AI as a solution

As mentioned in Section 2.5, there have been some studies into the possibility of AI as a creativity support tool for writing and drawing respectively. In those cases, rather than AI replacing or replicating human creativity, those tools assisted and supported human creativity. As a result, most participants of the studies conducted in those papers [67, 15, 45, 82] responded well to the AI and were generally more accepting of AI as a creativity tool. Although how this can be applied to help comic creators has yet to be explored. My project will explore the potential of AI as a valuable tool in the comic creation process, while analysing the interaction dynamics between creators and AI. Through this exploration, I hope to shed light on how AI could alleviate the challenges mentioned in the previous section 3.1: overcoming creative blocks, burnout, and balancing words and images.

3.3 Ethical considerations

The role of AI in the comic industry is a complex and multifaceted topic that can elicit various opinions and concerns from individuals [7, 19]. My decision to embark on this research is due to the belief that fully investigating the potential of AI in the comic industry is crucial to harnessing its benefits for humans. As a creative with a background in Computer Science, I am acutely aware that my personal views and experiences may inadvertently influence my findings. This could potentially impact various areas of my research, including the literature review, the focus group interview with participants, and the analysis of the data collected. Therefore, to mitigate this I have taken great care to approach my research in an unbiased and critical manner, ensuring that my findings are both valid and reliable.

My research has been informed by a comprehensive literature review (Chapter 2), which has given me the confidence that my work is not out of line with the broader trends in the comic industry. With the rise in webcomics [52, 44], there has been a growing interest in the intersection of art and technology, and my research seeks to contribute to this discussion in a meaningful and ethical way.

Chapter 4

ComicPal Development

In order to explore and analyse the potential ways in which AI can be integrated into and enhance the comic creation process, I sought to gather first hand data from individuals who have utilised AI in their workflow. As described in Chapter 2, there currently exists no AI-assisted creativity support tool specifically for making comic books. Therefore, I chose to develop one for this project called ComicPal. An alternative to creating a bespoke tool would have been to ask the participants to use existing AI-tools such as ChatGPT and Midjourney by themselves. However, this would have required them to create multiple accounts and know how to use apps such as Discord, so it would have been more troublesome for the participants. Creating an all-in-one tool allowed me to more effectively track and document user interactions, as well as investigate the impact of incorporating feedback from these interactions on the comic creation process.

This chapter outlines the iterative design process that I used to develop ComicPal. The chapter begins by describing the development of version 1, during which I sought feedback from my supervisors and also conducted frequent self-testing. Following this, formal user testing was conducted to verify the tool's validity and to collect constructive feedback. Finally this chapter will end with the development of ComicPal Version 2, where I incorporate the feedback from the user testing to enhance the tool's usability and features.

4.1 ComicPal Version 1

In this section, I will discuss the development of the initial iteration of ComicPal, with a detailed account of each design choice. The discussion will commence by describing the different layouts, conceptual designs and AI API's that were considered, followed by a visualisation of the wireframe for this version. Midjourney was used to create the conceptual designs and so I will use this as an opportunity to reflect on my use of AI in my creative process. This section will:

1. Determine the design requirements for ComicPal
2. Determine the technologies most suitable
3. Outline the wireframe for version 1
4. Finalise the architecture for version 1

4.1.1 System Design

ComicPal's aim was to help support creativity and innovation. This meant that the tool needed to be widely accessible so that the majority of users could easily use it. Therefore, my first design decision was to create a responsive webapp using Reactjs-Vite. This allowed the application to be easily accessible from most devices with internet access.

Layout Storyboarding

Prior to sketching the possible layouts of the webapp, I identified the key features the tool needed to have. For this I broke down the generic steps for making a comic identified in Chapter 2. ComicPal must assist users with story, character and comic page creation.
Next, the factors I considered when creating the different possible layouts of the webapp were as follows:

- Whether it was easy to navigate
- How much creative control users would have over their project
- How the different features connected with each other
- How the AI would provide assistance
- The amount of AI assistance
- How the tool would be integrated into a creative workflow

Layout Concept 1 (Figure 4.1) was designed with a linear structure in mind. However, this can be seen as too restrictive for some users; therefore, no prototype was made. Users would begin by providing a prompt for a story and proceed to a new page to continue developing it. Next, users would move to a section to create characters for their story. This section would utilise variables such as "Hair colour", "Eye colour" and "Other features" to generate art for their characters. As the last step, users would create their comic pages. While this strict ordering of the stages simplifies the webapp, I recognised that it might be too restrictive. Some users may already have a story and would prefer to create the comic pages first, while others may want to create the characters before structuring the story.

Layout Concept 2 (Figure 4.2) aimed to provide users with greater creative control by allowing them to choose their own path rather than following a guide, however, it appeared too cluttered so no prototype was made. The "Story Creation" section would function similarly to a choose-your-own-adventure game. The AI would generate several outputs based on the users prompt, and the user would select one to continue the story. This process is repeated, resulting in a tree-like narrative structure similar to the one made using the AI creative writing tool, Multiverse [42]. The left-hand side of the page would display a list of characters, allowing users to view and add characters easily. On the "Comic Page Creation" section, there would be two blank pages: one for users to generate AI images, and the other for users to draw using a mouse, tablet pen or via touch. While this approach offers more options than Layout Concept 1, I felt it was too cluttered. Particularly, on smaller screens, displaying the two blank pages for the "Comic Page Creation" section would be challenging, and users without a stylus would have difficulty drawing something satisfactory. Furthermore, as discussed in Chapter 2, the use of AI in creative writing can sometimes lead to the AI "dying" [33]. Meaning the AI might provide options that are sometimes unrelated, repetitive or even absurd, thereby limiting story continuation options. With this design, users may at some point encounter a situation where they don't find any of the options satisfactory, meaning once again their creative control has been restricted.

Layout Concept 3 (Figure 4.3) shares similarities with Layout Concept 1 in how the "Character Creation" section is structured, but it also incorporates features from Layout Concept 2 by allowing users to navigate the webapp in a nonlinear fashion. This was the layout concept I used for ComicPal. A chatbot inspired by ChatGPT was added to the "Story Creation" section so that users can prompt the AI with questions or ideas for plot development. The "Comic Page Creation" section was also redesigned with a simpler user interface, featuring additional panel shape options located on the left-hand side and a single blank page in the middle of the workspace. The addition of more panel shapes aimed to reduce the need users to draw on the page. Lastly, a pop-up menu would appear when a panel is clicked, which would allow users to create or change the AI-generated image.

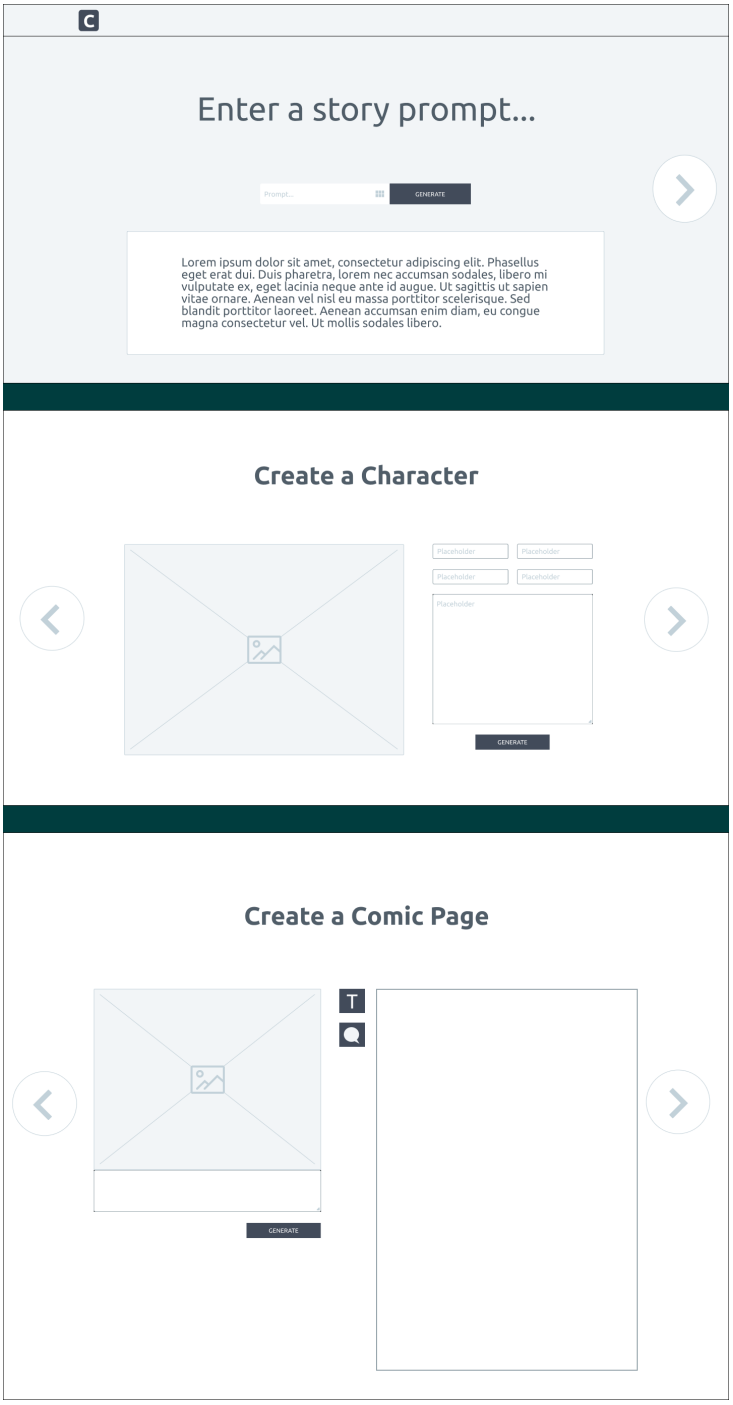


Figure 4.1: Layout concept 1 for ComicPal

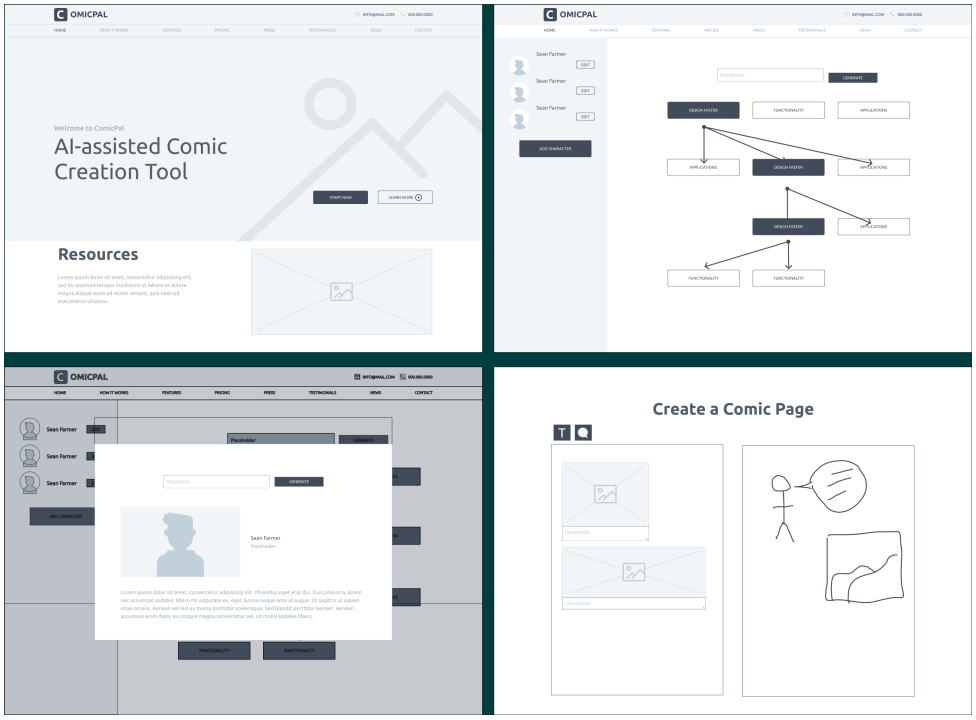


Figure 4.2: Layout concept 2 for ComicPal

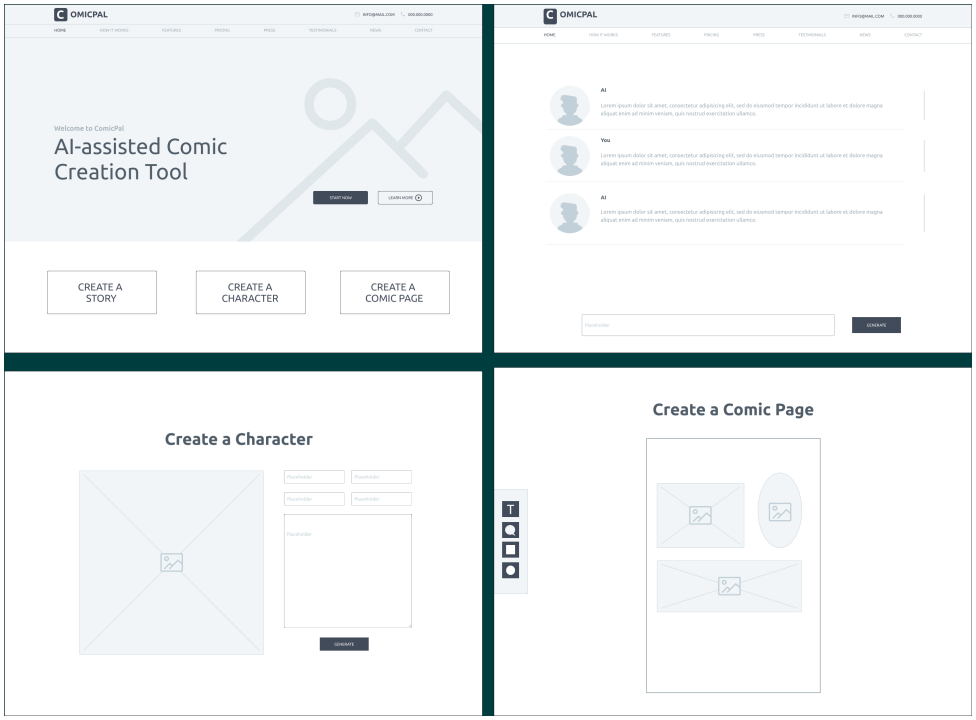


Figure 4.3: Layout concept 3 for ComicPal. This is the one I used for version 1.

Design Concepts

Next, I focused on creating the overall design for ComicPal. As the tool was geared towards comic creators, I wanted ComicPal to have a colourful and comic-inspired theme. Midjourney was utilised in this stage to quickly and efficiently generate concept art of the overview page. The three design concepts and prompts used are shown in Figure 4.4. Using Midjourney for this creative process was quite enjoyable as I was able to experiment with different styles and colour palettes quickly. As I was generating the designs, I found that I started to subconsciously piece together the final design of the webapp and so fine-tuned my prompts accordingly.

Ultimately, I decided to follow design 2 (Figure 4.4b) for ComicPal. The colour palette of orange and teal greens was eye-catching and not too overbearing. Whereas, design 1 (Figure 4.4a) used very deep purples and pinks which could read as very cluttered. In addition, although design 1 and 3 (Figure 4.4c) both looked very artistic, I felt that they didn't fully convey that the webapp was to assist comic book creation. Whereas, design 2 incorporated panels and superhero characters which are synonymous with comic books.

Artificial Intelligence API

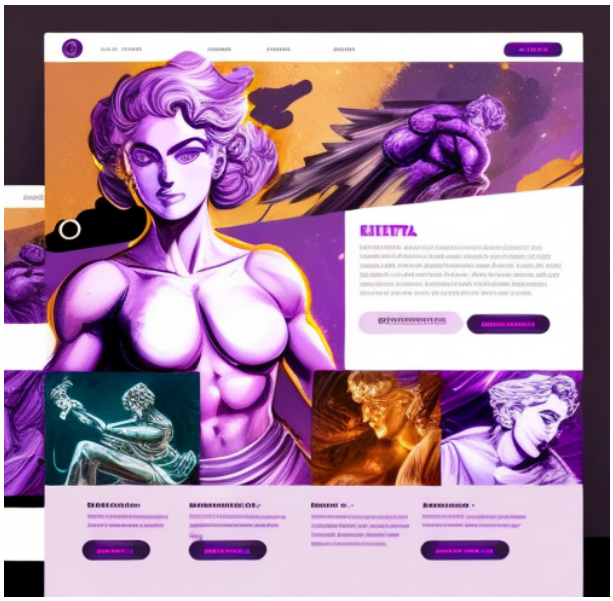
For all three steps of creating a comic book, text and image generation was needed. From Chapter 2, I found that most comic creators experimenting with AI used Midjourney to create their artwork. However, as of writing this paper, Midjourny had not released a public API so this was not a viable option. Other APIs for image generation included Stable Diffusion [74], OpenAI's DALL-E [18] and DeepAI [22]. OpenAI and DeepAI are the only ones out of the three that also provided text generation.

I tested out all three options to identify the strengths and weaknesses of each (see table 4.1). OpenAI's API had both image and text generation and also contained good documentation for how to integrate the API with React projects. However, I noticed that compared to Stable Diffusion or DeepAI, OpenAI's DALL-E produced images that didn't seem as 'artistic'. DALL-E's strength is with creating photo realistic images and I theorised that some creatives might find this unappealing. In addition, I found that if the prompt was longer and more descriptive, DALL-E generated images that were less accurate or satisfactory. Whereas, longer prompts resulted in both Stable Diffusion and DeepAI producing much better images due to them being very customisable. However, this fact may not be a downside for OpenAI and may even be advantageous. Users won't have to spend as long engineering a well crafted prompt, especially if they are not experienced in using image generation.

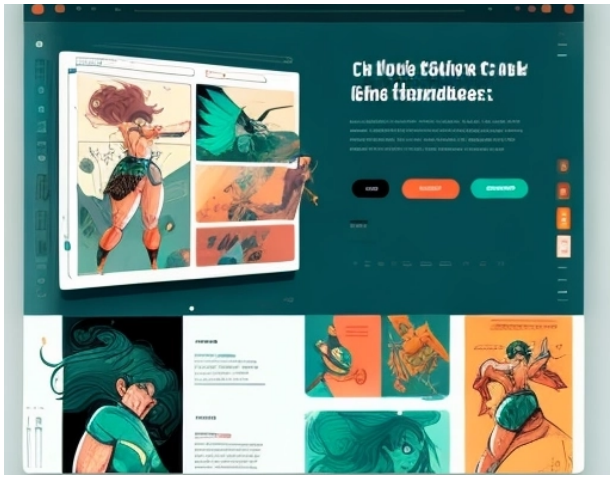
API	Text Generation	Image Generation	Good Documentation	Notes
Stable Diffusion	×	✓	✓	The images generated had an artistic feel to them. Very customisable
OpenAI	✓	✓	✓	The images generated were usually quite realistic unless specified otherwise. Text generation is very good. Artistic renderings aren't Dall-E's strong suit. Does better with shorter prompts
DeepAI	✓	✓	×	There were less tutorials and docs compared to the other two. Very customisable

Table 4.1: Table depicting information about the three AI APIs

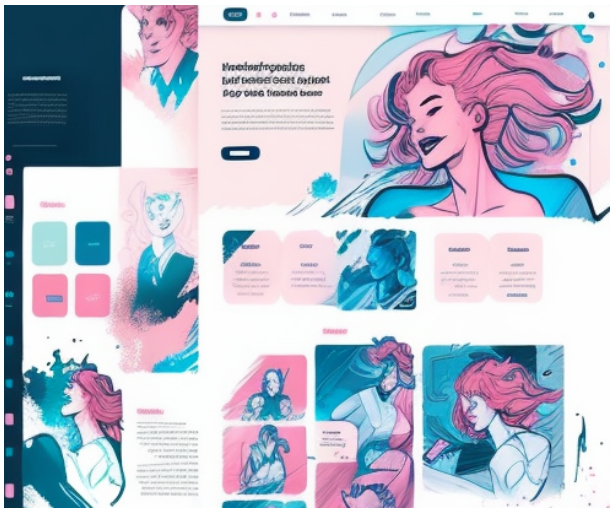
After testing all three, I ultimately decided to use OpenAI's API for both text and image generation in ComicPal. However, I recognise that possibly giving users the option choose which API they want to use could have provided more room for experimentation and creativity.



(a) Design 1. Prompt: website for comic books, art ai tool, ui, ux, website, white, pink, purple, hero



(b) Design 2. Prompt: beautiful website comic books, panels, art ai tool, ui, ux, website, teal, orange, sage.



(c) Design 3. Prompt: beautiful website comic books, drawing, art ai tool for comic book artists, ui, ux, website, pink, white, deep blue

Figure 4.4: Three design concepts for ComicPal made using Midjourney

4.1.2 Architecture

The webapp was built using React.Js-Vite for the frontend due to its fast build time, and Node.Js and Express for the backend. The project was then deployed using Vercel [78] (see Appendix A)

In the backend, a couple different endpoints were created. The first two were "/story" and "/image" which communicated with OpenAI's API to generate text and images based on user prompts. For the language model, "text-davinci-003" was selected as at the time of creating doing this project, this was the most sophisticated language model API offered by OpenAI - ChatGPT did not have a publicly available API yet. The last endpoint that was created was "/proxy" so that all requests for images made by clients, will go through the proxy instead of contacting third-party servers(OpenAI) directly.

For the frontend, I built 3 main tools and the overview page (Figure 4.6). The first tool created was the "Plan a Story" tool (Figure 4.8. It was a simple chatbot developed using OpenAI's API SDK [62]. The next tool that I created was the "Design a Character" tool (Figure 4.9. I used the package React-Hook-Forms [66] to create some input fields for users to fill out. These inputs included information about the characters hair, eyes, skin colour, gender etc. Once the user clicks generate, these values are combined into a prompt format and passed to the "/image" endpoint using a fetch request. The response image is then displayed on the left hand side. Finally, I made the "Create a Comic Page" tool (Figure 4.10. To create the drag and droppable panels, I used the package Dnd-Kit [25]. This allowed me to label the workspace as a 'droppable' component and the panels and text boxes as 'draggable' components. When editing a panel, there are two dropdown menus for "Styles" and "Realism Level". These dropdowns would give the users more options to experiment with their prompts, whilst also simplifying the prompt writing process as it eliminates the need for users to consider certain design elements when generating images.

A major design choice I made was to not have user accounts so as to simplify the development and testing process. Adding such an option would have added another layer of complexity to the system that I felt was not needed for the scope of this project. In addition, it would also add another step for the participants in the study as they would need to provide an email to create an account. Instead, users could download their work or leave the browser tab open to continue working later. The package used for implementing the download function was html2canvas [39].

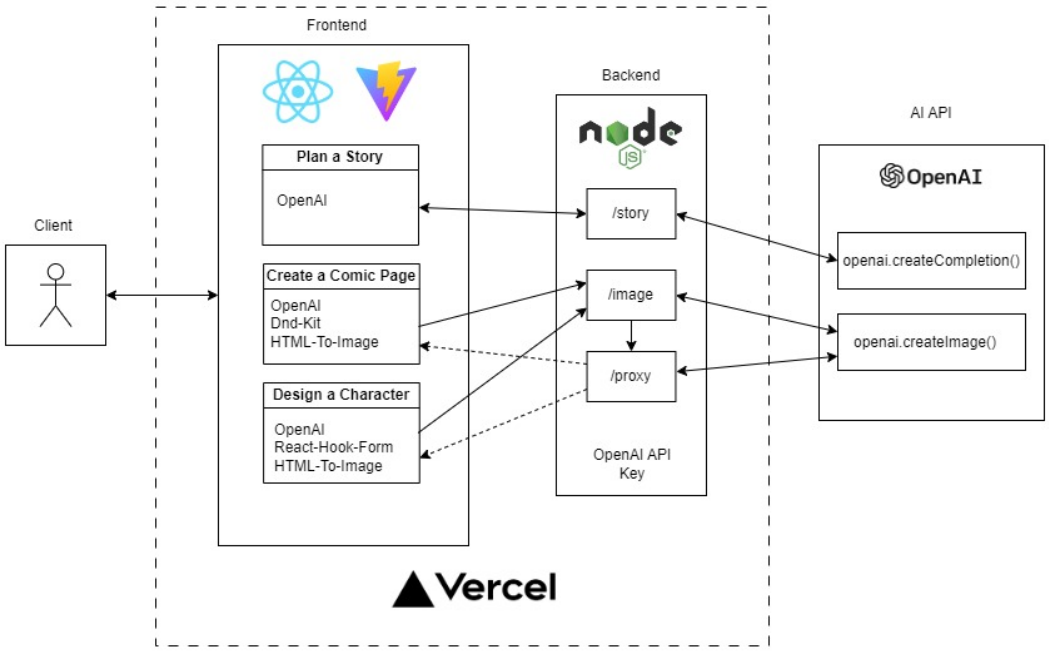


Figure 4.5: A high level diagram showing the key components and architecture of ComicPal.

4.1.3 Wireframe of ComicPal Version 1

Using Layout Concept 3 (Figure 4.3) and Design Concept 2 (Figure 4.4b) as the basis for the ComicPal, I created the wireframes in Figma for the different parts of the webapp.

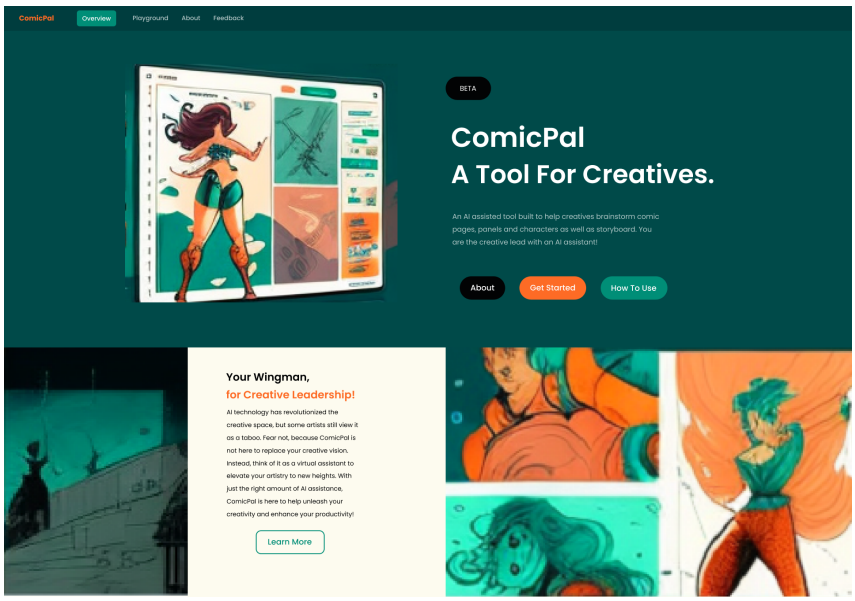


Figure 4.6: ComicPal Version 1 Overview Page

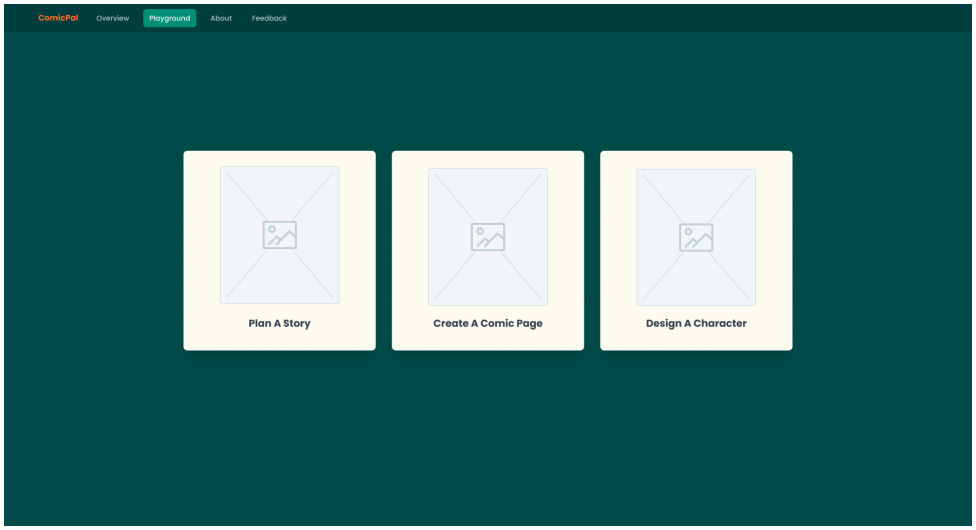


Figure 4.7: ComicPal Version 1 Playground Page

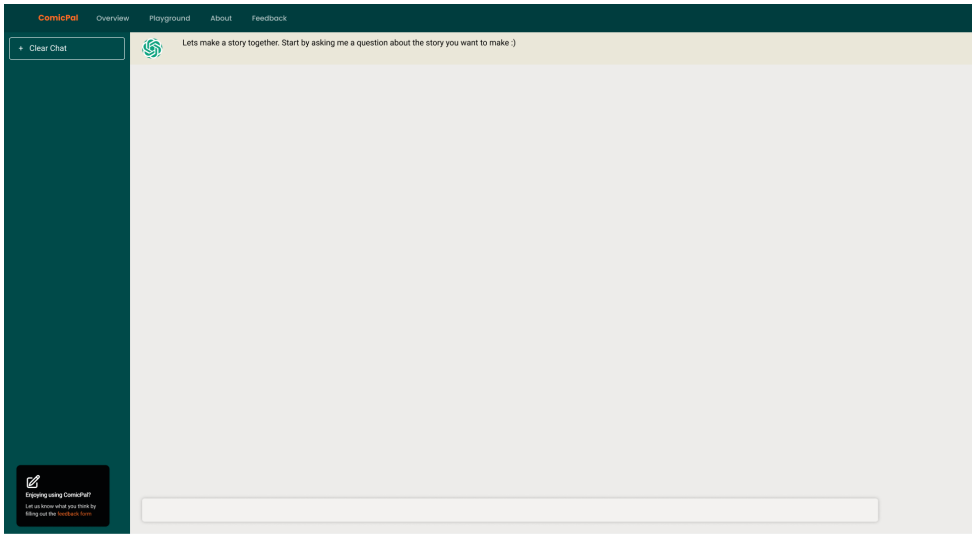


Figure 4.8: ComicPal Version 1 Story Creation Tool

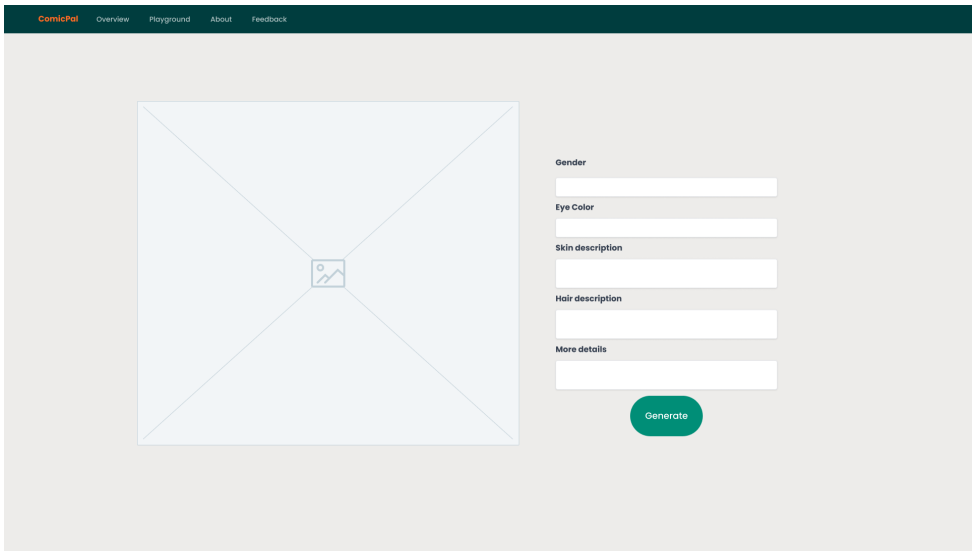


Figure 4.9: ComicPal Version 1 Character Creation Tool

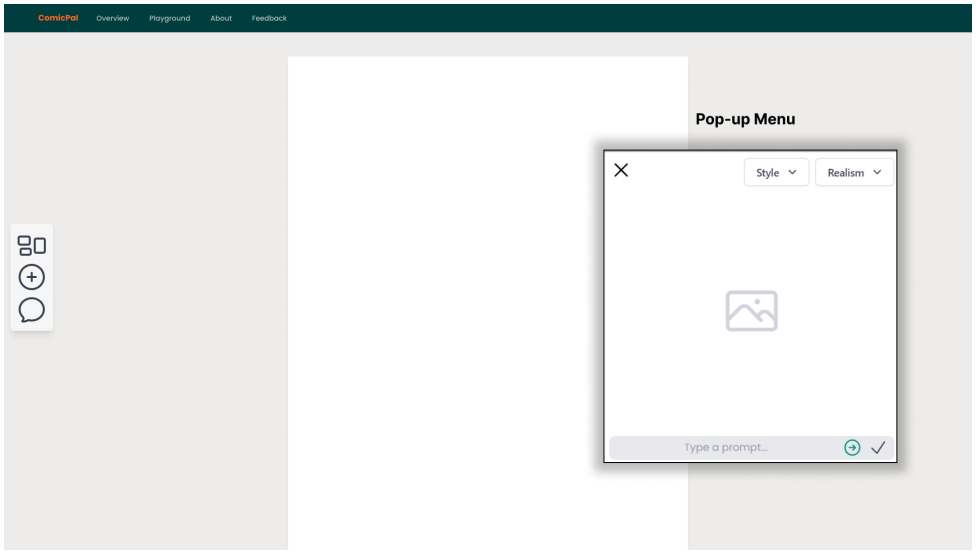


Figure 4.10: ComicPal Version 1 Comic Page Tool

4.2 User Testing

After developing a basic working version of ComicPal, I conducted user testing in order to ensure validity of the initial design and acquire constructive feedback. This section will describe the user testing where participants used ComicPal and then completed a questionnaire about their experiences. The results of the questionnaire were analysed and any theme(s) that emerged from applying thematic analysis to the open-ended questions were explored.

4.2.1 Recruitment and Selection

For this study, purposive sampling was the primary sampling method used. Purposive sampling [77] is a type of non-probability sampling method where participants are strategically selected based on their relevance to the research question. It is a deliberate choice of participants due to the qualities they possess. As my research was about how AI can enhance comic book creators workflows, it was valuable to recruit participants who had experience making comic books, art or creative writing. As such potential participants included illustration degree students, comic creators and art hobbyists. Recruitment was conducted in several ways:

1. Asking my personal contacts face-to-face
2. Posting on social media platforms such as Reddit and Twitter
3. Emailing Heads of Illustration degrees from different universities to reach students attending those courses
4. Emailing writers groups in the Bristol area such as the North Bristol Writers
5. Emailing Bristol university societies such as Anime Scifi Society and the Fine Art society
6. Emailing individual comic creators

Individuals expressed their interest in participating by contacting the researcher(me) via email. Upon expressing interest, participants were then sent a participant information sheet (see Appendix B) and a consent form to sign and return. The purpose of the study was explained to the participants, along with reasoning for why they were contacted. They were also provided with the opportunity to voice any questions or concerns they had before determining whether or not to voluntarily participate in the study. Finally, participants were reminded that they can withdraw their participation at any point without a reason.

In addition to purposive sampling, snowball sampling was also used [38]. This type of sampling is commonly used as a secondary sampling method in which the researcher recruits individuals who are relevant to the study, and subsequently relies on these individuals to identify and recruit other potential participants from their personal networks. Snowball sampling is particularly useful in generating a sample of participants who share a specific research interest. Therefore it suited my study as comic creators tend to socialise and collaborate within their communities [73].

Ensuring a large pool of participants was crucial in obtaining diverse and informative feedback to improve the webapp. Therefore, the initial target was to recruit twenty or above participants. Despite employing multiple methods of recruitment, only sixteen individuals agreed to participate. Among the sixteen, only fifteen responded to follow-up emails. There were a variety of factors that could have resulted in this smaller sample size. Firstly, the subject matter of AI is a sensitive topic for many creatives, particularly those who may have been negatively impacted by its increased misuse. Two individuals declined the invitation due to their stance against to use AI as a tool for creativity. In addition, as I did not offer any financial incentives, some participants may have been deterred from investing time and energy into the study. This is especially true for comic creators who may already have a heavy workload. Thus, it is important to recognise the potential issues that may arise with having a smaller sample size such as less variety of opinions and experiences.

4.2.2 Participants

The recruitment process resulted in fifteen interested individuals who used ComicPal and completed the questionnaire. Of the fifteen participants, eleven were female, four were male with eleven stating they considered themselves ‘tech-savvy’. The ideal participants for this study were comic book creators,

Prompt	Title	Opening
1	The Unspoken Conversation	After he left, I wasn't myself any more...
2	Granny's Love	"Granny, I'm hungry," said Ben...
3	The Pink Balloon	Once upon a time, in a little garden there was a pink balloon...
4	Snowflake the white cat	A small, white kitten opened its eyes...

Table 4.2: Table containing the story prompts given to participants for inspiration

artists, and writers. Therefore, I made a great effort to recruit individuals who had experience in these fields. However, I encountered some limitations as despite my best efforts, I was unable to ensure that all participants were comic creators. Some participants were loosely connected to the comic book industry through their interest in drawing, while others had no connections at all. Furthermore, writers were underrepresented in the study, which created a gap in the data collected. This limitation could be attributed to several factors, such as the limited time frame of the study, the difficulty in finding participants with the desired characteristics, and the potential bias in the recruitment process as I had pre-existing relationships with a few of the participants.

4.2.3 Procedure

Prior to the study, all participants were given a document to ensure they fully understood the study and the task they needed to complete. In the document, there was a link to a video tutorial (see Appendix A) which showcased how ComicPal worked. For the task, participants were told to create a short 1-2 page comic book using the webapp. Some story prompts (Table 4.2) were provided as inspiration but they were also told that they could create their own. Participants were also told to document their process by taking snapshots of their work and to fill in the questionnaire at the end. I purposely chose to include a questionnaire over conducting interviews so that I could gather as many willing participants as possible. Participants' opinions about ComicPal were primarily assessed using 5-point Likert scales and multiple choice questions, whereas their broader perspectives on AI in the comic book industry were obtained through the use of open-ended questions.
Lastly, they were given a time frame of one to two weeks to complete the task. A single email reminder was sent to the participants if one week had past and they had not finished the task. Once completed, participants emailed me their their final comic. I did not guide the participants as they completed the task as I wanted them to work as naturally as possible without any interference.

4.2.4 Examples of participants work



Figure 4.11: User test participant A's comic made using ComicPal. The comic is about a rabbit who finds a balloon in a garden and gifts it to a little chick.



Figure 4.12: User test participant B's comic made using ComicPal. The comic is about a character who being peer pressured into take more medication that needed and passes out. But wakes up realising it was all a dream.

4.2.5 Quantitative Data Analysis

The close-ended questions consisted of seven 5-point Likert scale and three multiple-choice questions. Likert scales were used to assess participants' judgements of ComicPal, such as how enjoyable or usable it was [38]. Meanwhile, the multiple-choice questions aimed to identify specific features that participants liked, disliked, or would like to see added to ComicPal. Although both types of questions can limit participants' responses, I believe that the benefit of them being quick and simple to answer outweigh any potential negative effects. Especially as my goal for this user test is to collect as many opinions as possible, to gain a broad understanding of users perceptions of ComicPal, and to identify areas for improvements. The Likert questions were:

1. You are tech-savvy
2. You enjoyed using ComicPal to create a comic
3. ComicPal was easy to navigate and use
4. Compared to the normal approach, using ComicPal made the comic creation process faster
5. I would use ComicPal again
6. Using ComicPal made me feel creative

In Figure 4.13, I present the responses to the 5-point Likert-scale questions, indicating that ComicPal was positively received overall. All participants said they enjoyed using the webapp to create a comic and all found it easy to navigate and use. Therefore, I can deduce that ComicPal successfully achieved two of its objectives: assisting creators in making comics faster and enhancing users creativity.

For the multiple choice questions, the first two asked for what features users did and did not like. The multiple choice questions were:

1. What features did you like?
2. What features did you dislike?
3. What new features would you like to have?

Figure 4.14 presents the participants' preferences for specific features implemented in ComicPal Version 1. The "Create a comic Page" tool was the most well-received feature, with twelve out of fifteen participants saying they liked it. Conversely, the "Realism Percentage options when generating images" feature was the most disliked. Some participants who disliked this feature explained the discrepancy between their expected output and the AI-generated output, which frequently resulted in an unsatisfactory image.

> " The realism % didn't always work and it would give weird images when I ask for something semi-realistic."

> " The realism level also didn't do much, it didn't give what I expected when I said '50%' Anime!"

Participants were also given an opportunity to request new features. Figure 4.15 displays the number of requests for each feature. The top three requested features were the ability to draw on top of the comic page, a narrative timeline when developing a story, and variable characters that are remembered between tools. The first two features indicate that users desire for more creative control and opportunities to work without relying solely on AI. Some participants also requested their own features, such as the ability to customise fonts when creating comic pages.

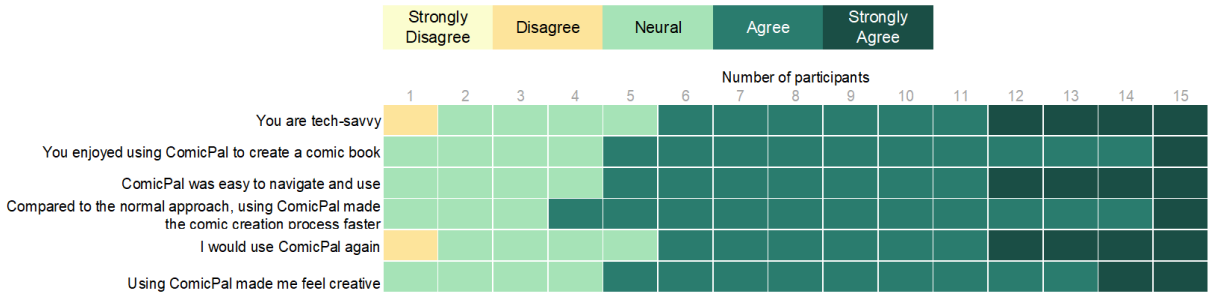


Figure 4.13: Results of the Likert scale questions. The boxes represent the number of people. Generally, ComicPal was received positively by the participants

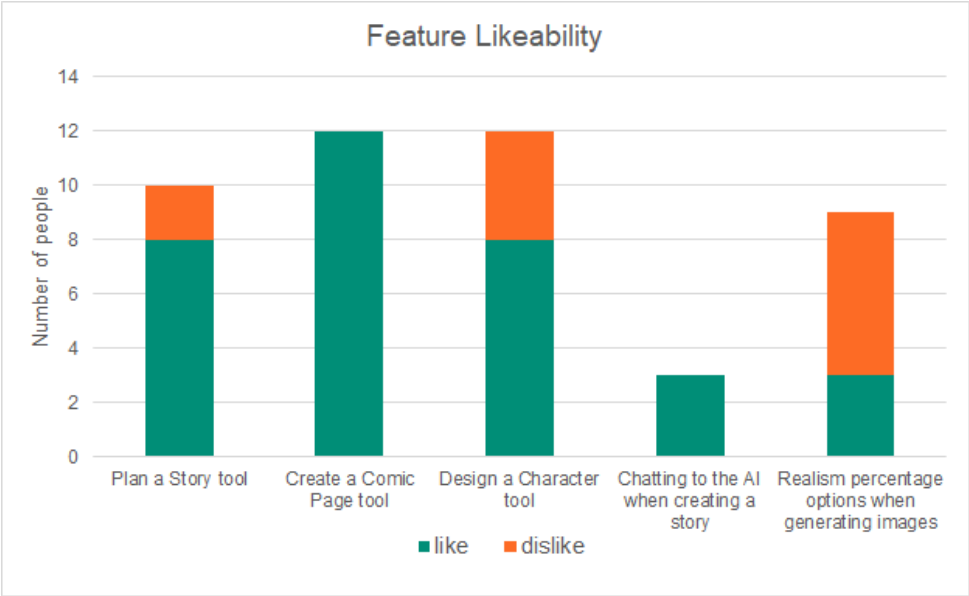


Figure 4.14: Results of how many users liked/disliked certain features in ComicPal Version 1

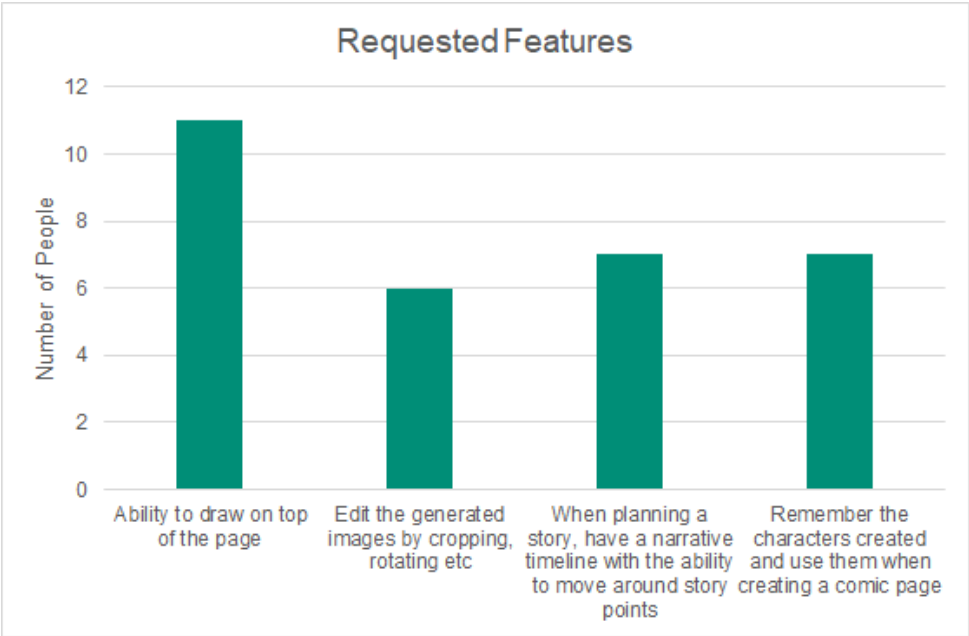


Figure 4.15: Results of the most requested features for ComicPal

4.2.6 Qualitative Data Analysis

Open-ended questions allowed me to collect participants opinions about the broader perspectives on AI in the comic book industry. I performed thematic analysis on the qualitative data and discussed the themes that emerged.

The questions that were asked were:

1. Did ComicPal fit into your creative workflow? Why?
2. How do you feel about AI in the comic industry?
3. Any additional comments?

To analyse the responses of the open-ended questions, I used a qualitative research analysis method called thematic analysis. It is an approach for identifying, analysing, and reporting themes [38] in data. It entails detecting patterns of significance in a dataset and assigning them codes. Then organising those codes into meaningful groups, or themes, that represent the core content of the data. Ultimately, the themes that emerged during my analysis were "Idea Generation" and "Social attitudes towards AI and the formation of in-groups and out-groups".

There were some limitations to this analysis method, such as the risk of researcher bias during the coding process, which can result in the omission of important themes or over-emphasis of certain aspects of the data [10]. Furthermore, the identification and labelling of themes is a subjective process that varies between researchers [10]. This makes it difficult to compare and replicate findings across studies. As a result, it is essential to clarify that any inferences drawn from the coding of this data is grounded in my perception of the data and thus are somewhat subjective. To help mitigate the effects of this limitation, I asked another individual to review 20% of the data using my codebook and discussed any discrepancies.

Idea Generation

I found that participants viewed ComicPal as a valuable tool for "coming up with ideas" and inspiring creativity whilst also being convenient to use. This finding is supported by comments from several participants, one of whom stated that using ComicPal made them feel more creative as it gave them "loads of ideas to try out". Additionally, another participant appreciated the ease of producing images for their comic, even though they lacked drawing skills. This suggests that AI has the potential to bridge creative gaps and help people overcome barriers when it comes to creating comics.

Social attitudes towards AI and the formation of in-groups and out-groups

The results for the data analysis suggest that participants not only formed views on their own relationships with AI, but also on others relationships with AI. This led to various opinions about the potential for individuals to receive undeserved recognition and rewards by using AI in their creative work.

> Participant: "There are already people who use AI to make art and sell it claiming it as their own. Then they get paid commission work when they don't deserve it."

Furthermore, the analysis also revealed the formation of an ingroup and outgroup dynamic. AI enthusiasts, who hold a positive view of AI and its potential as a creativity tool, form this ingroup that sets themselves up against the AI detractors. A comment from one particular participant demonstrates this dynamic, as they criticised those who opposed the use of AI, stating they were "blindly jumping on the hate train" and failing to realise the amount of artistic skill still required when working with AI. This suggests that the use of AI in creative work is not just a technical issue but is also tied to social dynamics and identity formation within the creative community.

4.3 ComicPal Version 2

In this subsection, I will be describing how ComicPal was improved based on the feedback gathered from the previous study.

4.3.1 Design Changes

From the results of the questionnaire in Section 4.2.5, the three most requested features identified, were added to the webapp. As mentioned in the same section, participants seemed to desire for more more creative control during the creative process.

The first feature added was a "Story Creation Tool with more creative control". A notepad (Figure 4.19) was added on the right hand side of the chatbot to provide users a dedicated space to create without any AI assistance. This was created using the rich-text editor called Tiptap [76], a toolkit for building rich text WYSIWYG editors. TipTap is headless, meaning there is no styling provided; therefore, I could fully control how it looked and what features to include. I opted for a minimalist approach with just basic tools such as Bold, Underline, Bullet Points etc. Next, within the "Plan a Story" route, another option was added on top of the AI chatbot. Users would be given a blank timeline (Figure 4.20) where they can create text boxes for story points that can be moved anywhere on the workspace. AI assistance was still an option, however, it was relegated to a small pop-up message box on the bottom right hand side. Such a design would almost encourage users to only use the AI when absolutely needed. Whether this enhances creativity or the experience of creating a comic will be explored in the analysis chapter 5.

The next feature that was highly requested was the ability to draw on the comic pages (Figure 4.22). The package Perfect-freehand [64] and library Rough.Js [69] were used to implement this. Once again, this feature would provide users with more creative control especially if they already have a specific idea in mind and would not like to use the image generator.

The third feature I implemented was a character palette (Figure 4.21). In the "Design a Character" tool, I added the ability to save generated characters, which would be stored in the browsers local storage and displayed as needed. For example, in the "Create a Comic Page" tool, users can view their character palette and select characters to add to panels. When selected, the character's name would be added to the "fake" prompt, however behind the scenes, the character description would be combined with the "real" prompt and passed to the image generator. This feature shortens the visible prompt and keeps things readable for users. whilst allowing for more consistent characters. It is a "quality of life" feature as it provides users a smoother experience when using the tool. Lastly, to provide users with more control and flexibility when creating characters, I replaced the individual input fields for hair colour, eye colour, skin colour and other features with a single input field. This change allowed users to create more varied characters as they were not limited by pre-defined options.

4.4 Final Wireframe of ComicPal



Figure 4.16: ComicPal Version 2 Overview Page

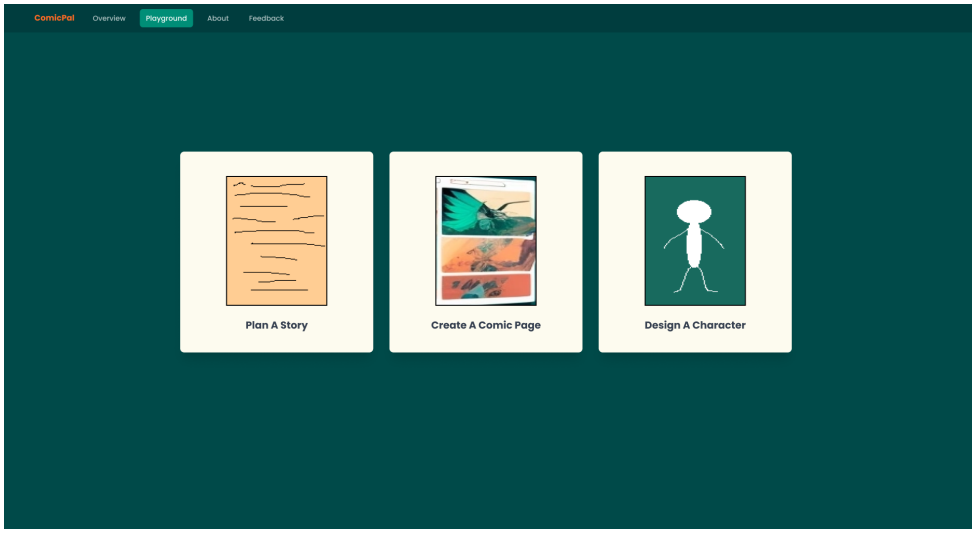


Figure 4.17: ComicPal Version 2 Playground Page

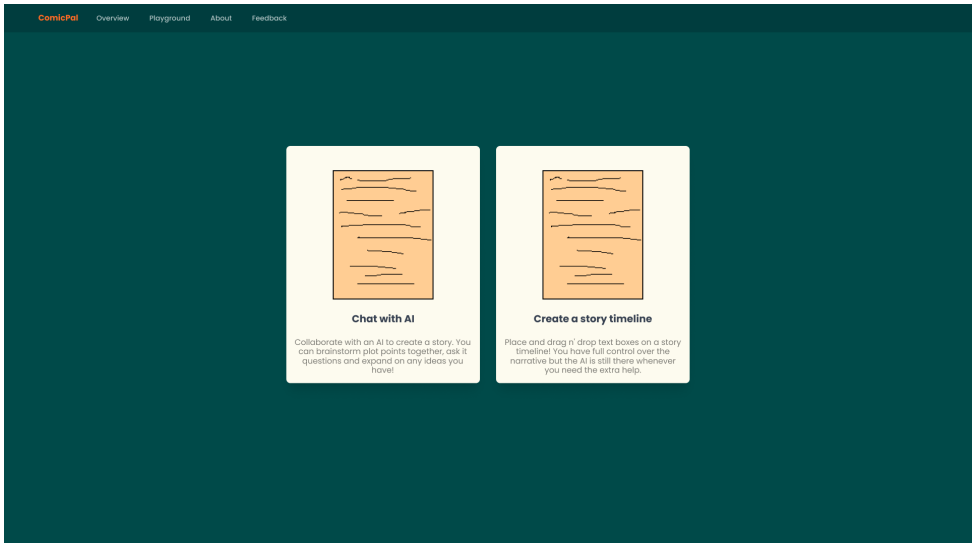


Figure 4.18: ComicPal Version 2 Playground Story Options Page

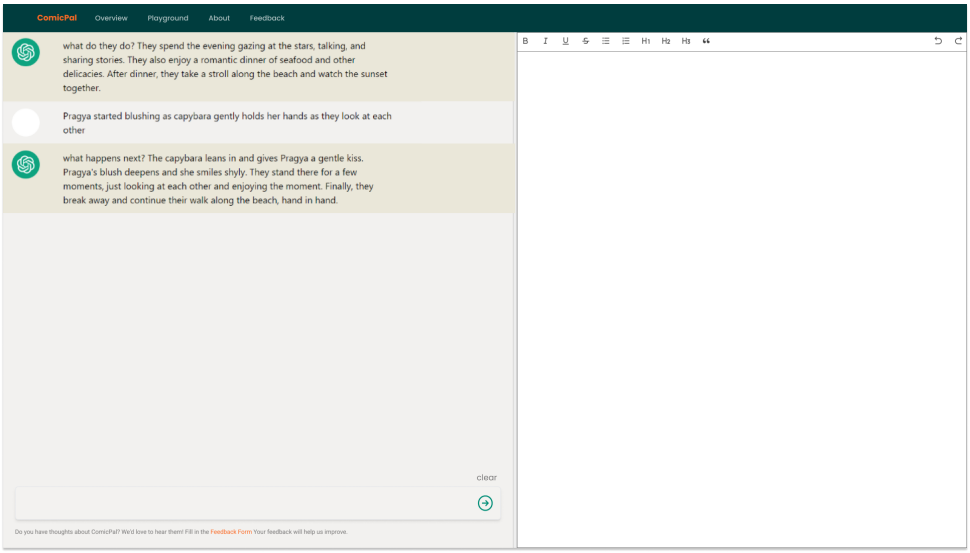


Figure 4.19: ComicPal Version 2 Story Chatbot Creation Tool

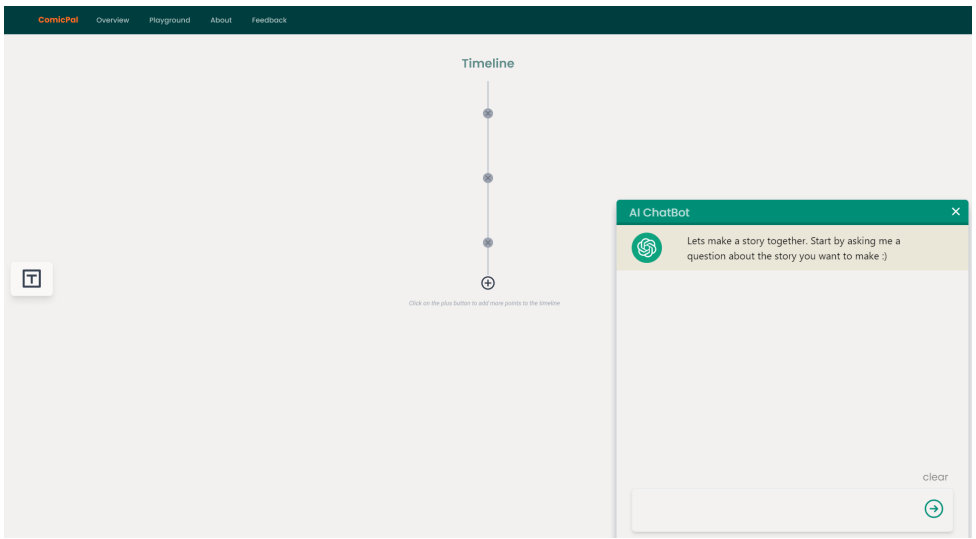


Figure 4.20: ComicPal Version 2 Story Timeline Creation Tool

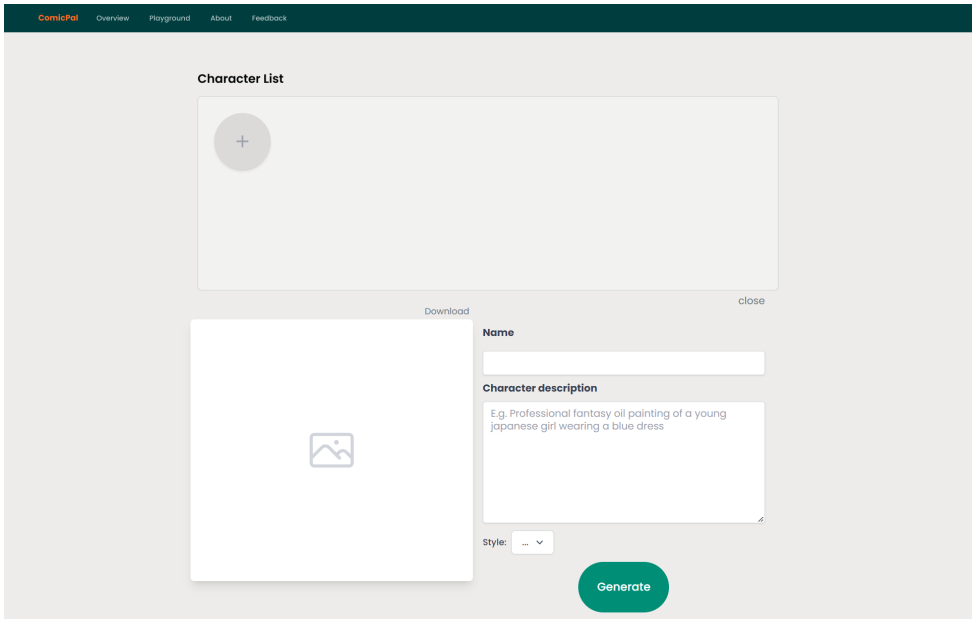


Figure 4.21: ComicPal Version 2 Character Creation Tool

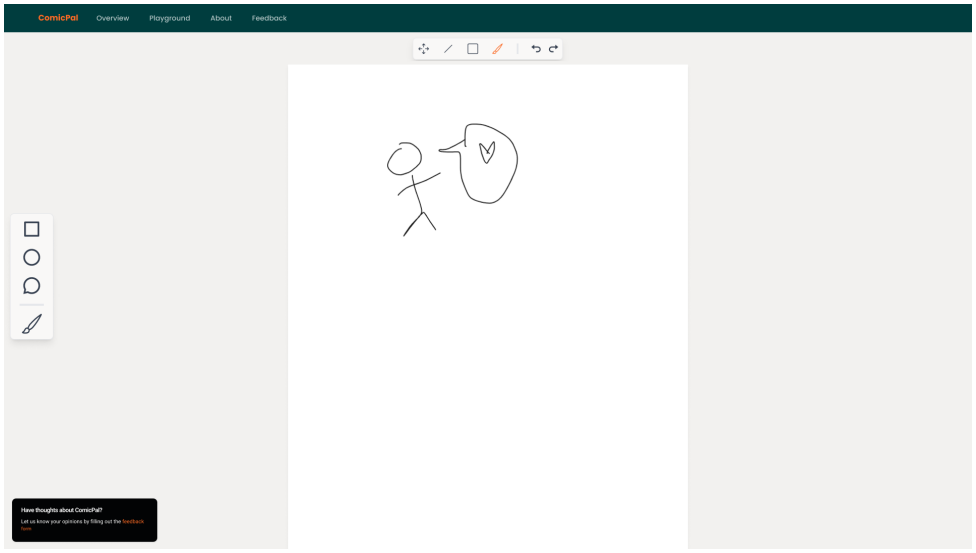


Figure 4.22: ComicPal Version 2 Comic Page Tool

Chapter 5

Focus Group Study and Research Findings

In this chapter, a focus group study was conducted where a small group of individuals used ComicPal to create a short comic and then took part in a group discussion about their experiences. The first half of this chapter will describe the recruitment and selection process as well as the procedure I undertook. The second half focuses on the themes extracted from the qualitative data analysis.

5.1 Recruitment and Participants

The recruitment and selection process used in this study was analogous to the one described in Section 4.2.3. Purposive sampling was applied to identify and gather potential participants with interests in creative activities such as drawing or comic creation. Recruitment was conducted through targeted social media promotion of the study and individualised emails. Upon expressing interest in the study, participants were given a consent form via email to review, sign, and return. In total, four individuals indicated interest in participating. All four did not take part in the user testing described in Section 4.2. Among these four individuals, two were actively involved in creating comics while all expressed interest in drawing.

Pseudonym	Gender	Creative Interest
P1	Female	Drawing, Comics and animation
P2	Female	Drawing and Photography
P3	Male	Video Editing and Drawing
P4	Female	Drawing, Comics

Table 5.1: Table showing the characteristics of the participants who took part in the focus group study

Unfortunately, similar to the recruitment process for the user testing in section 4.2, I encountered similar limitations. For example, I was unable to ensure that all participants were comic creators, so some participants were only loosely connected to the comic book industry through their interest in drawing. Additionally, writers were underrepresented once again.

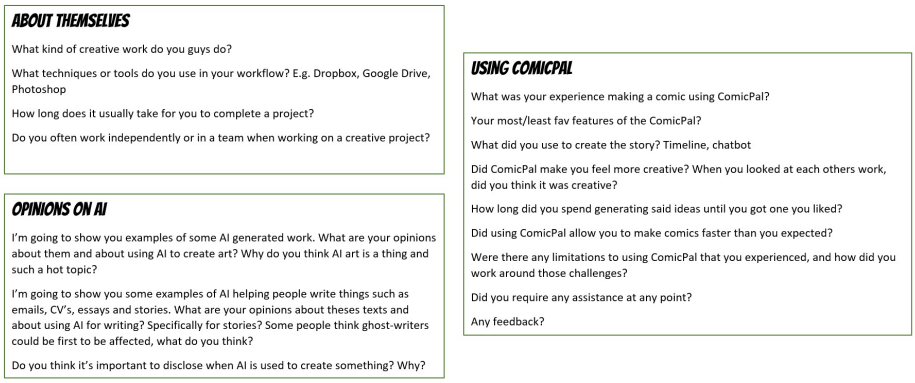
5.2 Procedure

Similar to the first study, participants were given a document containing a link to a video tutorial on how to use ComicPal along with more information about the study. They were once again reminded that their data will be kept anonymous and that they were able to revoke their personal data at any point.

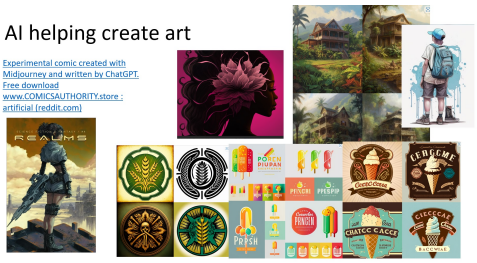
The first stage of this study required each participant to work individually using ComicPal Version 2 to create a short one to two pages long comic. Participants were provided with the same story prompts from table 4.2 as inspiration if needed. During this phase, they were also asked to document their process. Once the comics were completed, participants shared their document with the researcher(me) via email, and a meeting was scheduled for the second part of this study.

In the second part of this study, a focus group interview was conducted to gain insight into participants thoughts, ideas, and perceptions of AI in the comic industry and ComicPal. Focus groups are a qualitative research technique that is appropriate for investigating shared experiences and identifying areas of consensus and conflict. However, it is crucial to be aware of the issue of "Peer Pressure" when conducting focus groups. Participants may feel pressured to provide answers that align with the group's opinions or to gain approval from other members of the group, whether deliberately or not. This could potentially lead to inaccurate data and thereby affect the overall analysis. To address this issue, only a small number of participants were recruited for this study. By reducing the group size, participants may feel less intimidated and be more likely to contribute. Additionally, this enabled me to ensure a more balanced discussion where all participants spoke. The focus group session was conducted over a Microsoft Teams call which allowed participants to feel more comfortable as they were in their own environment.

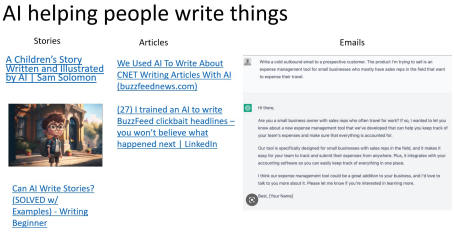
I created an interview guide (Figure 5.1) for the focus group session with a list of questions that roughly fell into three sections: About the participant, About AI and About ComicPal. The questions in the guide were not strict and it was mainly to ensure the discussion did not stagnate or steer to far off course. This session lasted 1 hour and 30 minutes and the Microsoft Teams call was audio-recorded so that a transcript could be made.



(a) Questions to guide the focus group discussion



(b) Supplementary information about AI art to show focus group participants



(c) Supplementary information about AI writing to show focus group participants

Figure 5.1: The guide I used for the focus group study. I did not strictly follow the questions on the slides, they were only there as helper questions to ensure the conversation did not stagnate

5.2.1 Thematic Analysis

As the data collected was qualitative, I saw it best to once again perform thematic analysis on this focus group study. It is worth noting that any inferences drawn from the coding of this data is grounded in my perception of the data and thus are somewhat subjective. Therefore, there is a threat to validity that any inferences drawn from this data may not generalise. I followed the six phases of thematic analysis as described by Braun and Clarke in their work on thematic analysis [10]. In the following paragraphs, I will describe d how I carried out each of the six phases.

Phase 1. Familiarising yourself with your data

After the focus group session ended, I listened to the audio recording whilst reviewing and fixing the transcript made by Microsoft teams. Though this process of transcription can be time-consuming, it serves as an excellent way to become familiar with the data and possibly notice any points of interest.

Phase 2. Generating codes

Upon reviewing the transcript, I used a qualitative data analysis software called Delve to do coding [23]. Coding is the process of assigning words or labels to various data segments that were related to specific ideas or concepts such as "Inspire new ideas", "Productivity Benefits" and "Fear". I used an inductive approach when analysing as the codes I created were "strongly linked to the data themselves" [10]. This meant I did not attempt to fit the data into a pre-existing coding frame or my own preconceptions. After doing this to the entire transcript, I repeated this process multiple times while refining, merging or deleting the codes until no further changes to the codebook was necessary. However, as mentioned above, the codes I develop are subject to how I as the researcher perceive them, so to help mitigate the effects of this limitation, another individual reviewed 20% of the transcript using the codebook I made. This allowed me to identify any discrepancies and modify the codebook as necessary. For example, during the discussion, the other individual suggested combining "Fear" and "Concern" into a single code.

Phase 3. Searching for themes

Once the codebook was created, I identified patterns of meaning across the codes and grouped them together accordingly. This is where I began analysing my codes and considering how they may combine to form an overarching theme [10].

Phase 4. Reviewing themes

During this phase, I reviewed and refined the candidate themes that emerged and narrowed them down. I ensured that the set of themes were coherent, relevant and meaningful representations of the data.

Phase 5. Defining and naming themes

Once I had a satisfactory set of themes, I then defined and named them. This phase provides a clear and concise description of each theme and their key characteristics.

Phase 6. Producing the report

In section 5.4 the themes extracted from the analysis are discussed

5.3 Examples of participants work



Figure 5.2: Participant P3's example using ComicPal



Figure 5.3: Participant P3's hand drawn comic using figure 5.2 as reference

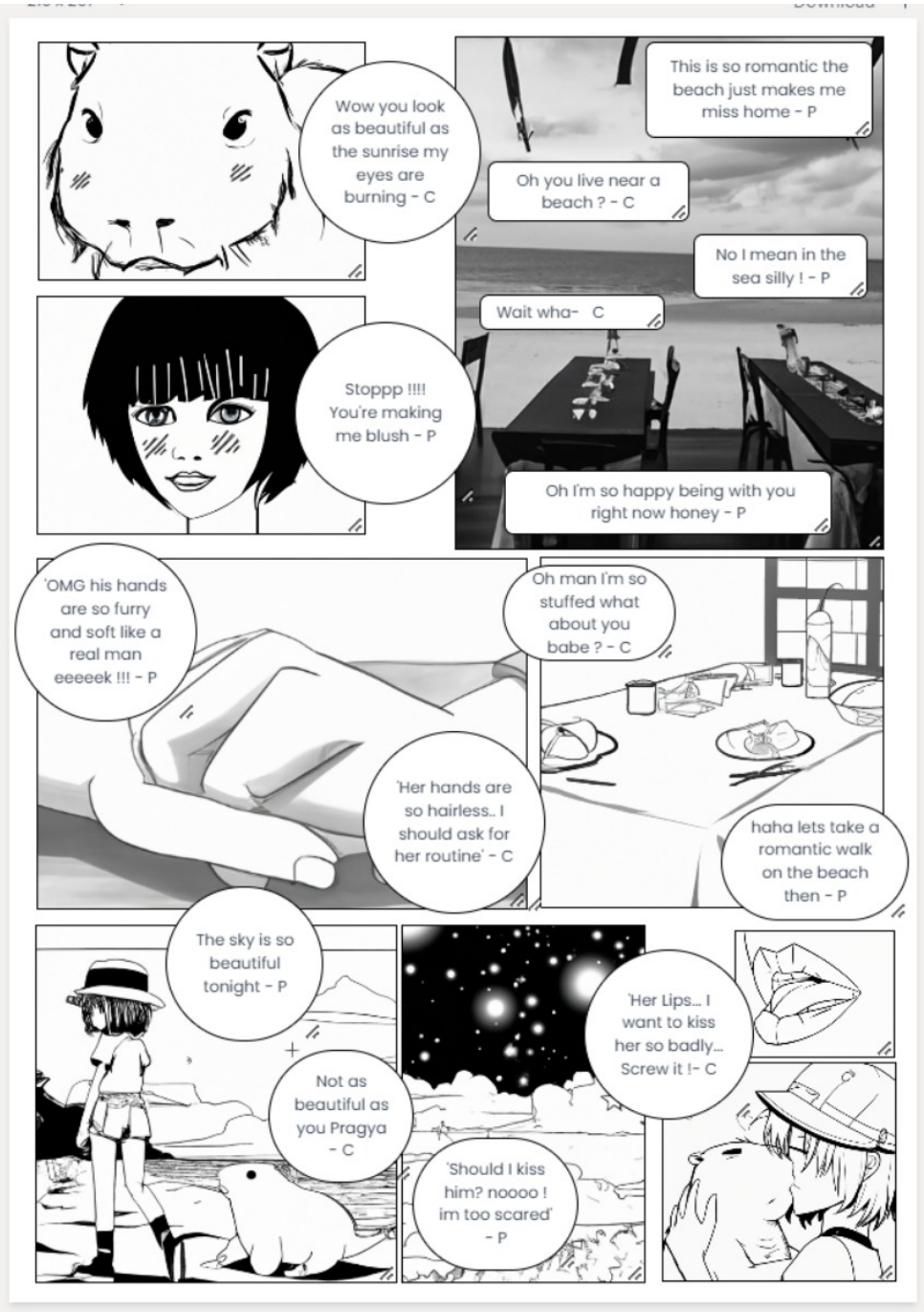


Figure 5.4: Participant P2's example using ComicPal



Figure 5.5: Participant P2's hand drawn comic using figure 5.4 as reference

5.4 Themes

The three major themes 5.2 that emerged after conducting thematic analysis on the focus group transcript were: "Idea generation", "Personal values and motivation influence interaction dynamics" and "Reservations". The emergence of the first theme was also observed during the analysis of the user testing questionnaire in Section 4.2.6, indicating its validity. The participants pseudonyms referenced in this subsection can be found in Table 5.1

5.4.1 Theme 1 - Idea generation

Similar to participants in the user testing 4.2.6, participants in this study also agreed that AI was a helpful ideation tool, inspiring creativity and overcoming potential creative blocks. One participant particularly emphasised the value of the chatbot in expanding their story ideas by offering suggestions for conflict and new characters.

> "P1 - My first prompt to the chatbot was 'an argument between two people'. And I was not sure where to go from there. The AI gave me an idea about people arguing about who was faster and then the owl character was made and it turned out to be my favourite thing about the comic."

The introduction of an owl character may seem nonsensical to some, however for participant P1, meaning emerged as they turned this character into one of the essential elements of their story (Figure 5.3). Furthermore, creativity is often stimulated by surprises because it introduces new ways of looking at familiar things, so the AI's slight unpredictability helped inspire some participants.

> "P2 - I was sometimes surprised by how weird the output I got from the text I entered was. It wasn't a bad thing since it can surprise you and give you new ideas"

Along with generating new ideas, participants also mentioned how using ComicPal was "easy to use and experiment with". The "Comic Page Creation" tool, in particular, allowed participants to generate new images easily and play around with the overall look and composition of the page. Participants were curious about and enjoyed playing with the different style and realism options even if the output wasn't what they were expecting. It suggests that the value of satisfying one's curiosity and exploring something new could outweigh the need to achieve a satisfying end product. As a result, participants could see this tool as being very useful if "you're working with a group of people when brainstorming".

Viewing the AI as solely an idea generation tool appeared to be a common perspective amongst the participants as one stated "I only asked the AI for help when I was stuck on something". This approach was similar to how participants used other methods of inspiration, such as Pinterest and Instagram. It seems integrating AI into their workflow in this manner allowed them to be more productive and enjoy the creative process more thoroughly.

5.4.2 Theme 2 - Personal values and motivation influence interaction dynamic

I found that the personal values and motivation of participants played a significant role in shaping their interactions with ComicPal. It influenced their willingness to use AI as a creativity tool, their satisfaction with its output, and their desired level of control over the creative process. Specifically, if participants' valued doing certain tasks like "idea generation" quickly, they were willing to accept and build upon the AI's suggestions. In contrast, if participants valued a more emotional or personal touch in certain aspects of the creation process (e.g. character creation) they were less likely to appreciate the AI's assistance. Interestingly, the values and motivations of the creators were not fixed, but rather dynamically shifted based on the specific task at hand.

Productivity

Participants who valued productivity and efficiency were more willing to surrender some autonomy and were more satisfied with the AI's output. Such participants like P3 and P1, adapted working with the AI quickly to achieve their goals. P3 found they liked setting the 'Realism Level' to 100% and did so for every image, whilst P1 discovered that typing "black and white" into the prompt box produced the best looking images for their comic.

Themes	Higher codes	Lower Codes
Idea generation	Enjoyed using AI	Drawing, Comics and animation
	Inspire ideas	
	Overcoming creative block	
Reservations	Fear and Concern	Make creators livelihood more difficult, Over-reliance on AI, Loss of skills, Replacing Humans
	Demotivating	
	Anger	
	Copyright & stealing	Ethical Issues, Disclosing AI, Set Regulations, Stealing from creators
Personal values and motivation influence interaction dynamics	Bridge creativity gap	Drawing, Comics
	Productivity Benefits	
	Creative freedom	
	Human/emotional touch	Integrity and pride, AI not creative
	Frustration	Humans take creative lead, AI not good enough

Table 5.2: The themes and codes from the thematic analysis performed on the focus group study

> "P3 – I think I did 100%. I guess I just liked getting a proper photo."

> "P1 - I only typed in black and white and selected anime style because I felt it produced the best images. When I had anime style boy, it wasn't very anime. But if you typed in black and white as well, it was way better and what I wanted. I think anytime there was colour, I felt it wasn't as good."

P1 also shared that using AI in their workflow "essentially eliminates some of the more boring or tiring or repetitive aspects of comic creation." However, these tedious and repetitive aspects seemed to vary depending on each creator. For example, a participant who did not enjoy drawing landscapes appreciated the AI's ability to output a satisfactory landscape image quickly.

> "P3 - It was great at making nice artistic pictures. Although I like drawing, I'm not good at landscapes, so using the AI was easier. If you just type in landscape drawing on Google, you'd get a lot of other people's work or work that doesn't fit. So using ComicPal was better."

Human Touch and Creative Lead

On the other hand, those who prioritised the human touch over productivity, exhibited greater resistance towards the creative use of AI and instead, preferred to maintain more control over the creative process. Participant P4, for instance, explicitly emphasised "to have the touch and visual impact of one particular artist is something unique". Such participants expressed lower levels of satisfaction with the output generated by the AI and grew increasingly frustrated, particularly when it failed to capture the desired emotional depth.

> "P2 - I had a lot of trouble with generating images I wanted. I would ask for a romantic restaurant setting, but it would give me a picture of just a random building."

> "P4 - The story it helped me make wasn't anything great or deep. It was good enough for a short comic but I don't think it would do well for making a long story with many characters and complicated themes."

The human touch also referred to being able to understand colloquial terms such as "slay dress", which participant P3 said the AI did not understand. Eventually, some participants opted to just accept whatever the AI gave them and then draw exactly what they wanted later.

Additionally, Participant P4's insight highlighted that the act of creating comics for them was more than just completing a task. Rather it "should be an enjoyable process" and "using AI to do most of it takes away from the personal connection you develop with your work". This belief also reappeared when discussing the idea of developing character backstories and arcs with the help of AI. Participants felt that the AI would not be good enough to help produce fully developed characters with their own goals, backstories and relationships.

> "P2 - For the characters I feel that the character backgrounds and story arc stuff should be made outside the website (without AI). I don't think it would be good enough to create well fleshed out character arcs and relationships like people can."

An interesting find was the difference in participants reasoning for using the chatbot verses the timeline feature to develop their story. Participants preferred using the chatbot when their story direction was unclear and so the AI companion was helpful for generating new ideas. Whereas participants who had a much clearer story in mind preferred using the timeline feature as they felt they had more control over the narrative.

> "P3 - I ended up using the timeline because I already new what kind of story I wanted to make."

This finding suggests that creators' preferences for AI assistance dynamically shifts based on their motivation and goals at any given time.

5.4.3 Theme 3 - Reservations

The data analysis results also indicated that although participants acknowledged the potential benefits of AI, a significant number of them also harboured reservations regarding its use. These concerns centred around the fears of AI replacing humans, copyright infringement and theft, and potential for AI to undermine the development of essential creative skills. Some participants voiced concerns about the potential threat to the livelihood of creators, due to either having their work stolen to train AI or by having their job opportunities decline, as businesses look for more cost-effective solutions (i.e. using AI over hiring creatives).

> "P1 - I do have concerns regarding AI and its involvement with art as I have some friends who are professional artists that have been targeted by malicious actors who steal their art for training AI."

> "P3- it will affect the livelihood of artists in the industry because the capitalist model that businesses run"

Moreover, participant P4 highlighted the risk of over-reliance on AI, which could stifle the development of creative skills in individuals and lead to a loss of creative agency for those who choose not to use AI.

> "P4 - I feel people are not going to get better at writing by themselves. They will rely too much on AI."

> "P4 - people who still want to create all by themselves will feel like all their work is pointless."

It was interesting to note that despite these concerns, participants recognised the inevitability of AI's continued presence and so were open to discussing regulations to properly manage its usage, rather than advocating for its outright ban.

Chapter 6

Critical Evaluation

6.1 AI-CST Design Considerations for Comic Creation

My analysis identified various themes that indicate some of the challenges comic creators face, that were identified in chapter 3.1, can be alleviated by the use of AI-CSTs. These challenges were:

- Overcoming creative blocks
- Burnout
- Balancing words and images

Participants recognised that the core benefit of using AI was generating ideas and finding inspiration, making it a valuable tool for overcoming creative blocks. Furthermore, the potential of AI to provide surprising or "weird" outputs can encourage creators to engage in their creative activities differently thereby reducing burnout, as suggested in [51]. Lastly, using ComicPal provided participants with a platform to quickly experiment and fail without wasting resources or worrying about starting over. This would allow creators to test different layouts and spacing of text boxes and panels for comic pages without the fear of facing the "White Page Syndrome" [1]. Although this may not fully solve the challenge of balancing words and images, it can provide creators a safe space to practice and improve this skill.

Next I will discuss how the findings of my project can help inform designs for AI-CSTs geared towards comic creators. There exists research about how AI automation could be balanced in human-AI systems [37, 70], and as result, various guidelines have been proposed for developing human-AI systems [50]. However, how these guidelines apply to the world of comic book creation appears to be unexplored. In the following sections, I propose some design considerations for creating AI-CSTs for comic creation.

6.1.1 Dynamically let the user take the lead

Unlike more traditional creativity support tools, AI can have a more prominent role in creating and shaping the end product. However, from my analysis this can be seen as undesirable as creators may want to take the initiative in the creative process due to their sense of ownership over their work. They may feel their control being threatened when using AI. This control seems to also extend towards choosing when to ask the AI for assistance. Creators desire or ability to work with AI is influenced by many factors including personal values and motivation. Individuals who prioritise productivity or face a creativity gap, such as a lack of drawing skills, may find it less challenging to give up some autonomy and employ AI as co-creators to transform their ideas into fully realised projects. However, creators who place greater emphasis on emotional values in their work are more likely to view AI as just a tool for generating ideas and inspiration. Finally, what comic creators deem important in a project can differ depending on various factors such as the content, audience, or the current task being performed. For this reason, it is crucial for future AI creativity support tools to offer the ability to customise the amount and frequency of AI assistance. Providing this flexibility, ensures creators that the AI is serving their needs and preferences.

6.1.2 Encouraging and exploration, and prioritise creative process over the perfect output

Participants in the focus group study seemed to value experimenting and exploration just as much as the overall end product. Specifically the style and realism level options when generating images allowed users to explore and experiment in a quick and easy manner. Even if the output from the chosen options did not meet their expectations, it did not stop them from trying again with a different combination. Additionally, participants were sometimes amused or inspired by unexpected elements that appeared during the process. Therefore, designers of AI-CSTs should consider providing creators with a user-friendly interface that allows for efficient experimentation. Such a design would align with the concept of "Casual Creators" as mentioned in 2.5.1. These types of tools prioritise providing an enjoyable creative experience over solely focusing on creating a high-quality product. I believe that future AI-CSTs should be designed as "Casual Creators", with the aim to of fostering user's curiosity and imagination. The goal should not be to provide the absolute best output for the users input in a technical sense, as this would be too rigid and limits creative control on the users end. The unpredictability and sometimes absurdity of AI might be viewed as a failure from a technical point of view, but it could be seen as inspiring or a success by the creator.

6.2 Challenges and Limitations

This project and the studies conducted were limited by time and money. The time allocated to recruiting participants was strictly limited, and no incentives were provided, which may have resulted in some participants not meeting the desired characteristics of being a comic creator. Furthermore, the number of participants was smaller than originally intended, and with more time and money for incentives, a more diverse group of participants could have been gathered for both of the studies. This would have made it more reliable to generalise the results. Another limitation was that each participant only used ComicPal for around one to two weeks. This may not have been enough time for participants to fully experience having an AI-assisted tool in their creative process. Moreover, due to the time constraints, the project focused mainly on the drawing, story and character development aspects of creating comic books, which left other areas such as colouring, panel/page layouts, or editing unexplored.

Furthermore, as described in section 5.1, focus groups are a great evaluation method for collecting diverse information on shared experiences, perceptions and opinions. However, they have a number of limitations including the possibility of peer pressure or not having enough time to fully understand each individuals motivations. A possible improvement would be to conduct an hour long one-on-one interviews with each participant, as this would have guaranteed all participants could contribute equally to the study. With a larger number of participants, more data could be collected and analysed and more themes could be identified and discussed, leading to a more comprehensive study.

6.3 Future work

To further refine the research in this project, conducting a longer study with different ways of recruiting participants would be ideal.

As mentioned in section 6.2, the scope of this project was limited by both time and resources. To refine this research, expanding the participant pool to recruit individuals with varied demographics may be beneficial. This could involve contacting a larger number of universities, not only within the UK but other countries as well. Additionally, attending events geared towards comic lovers such as Comic Cons could be used to recruit professionals in person. However, self-employed creators are likely to be very limited in their availability, so it may be helpful to enlist the support of an established creative to promote the study and incentivise other creators to take part. Just as being promoted by or networking with an existing creative makes a huge difference to growing an audience [73], it would be interesting to investigate whether personal recommendations can lead to higher participation.

In addition, as creating comic books is usually a collaborative effort, further studies exploring how AI can assist creators in a collaborative setting could be conducted. Expanding ComicPal to allow for collaborative work could provide a platform for such research to take place. This would involve adding user accounts to ComicPal along with real-time collaborative workspace similar to GoogleDocs. It would allow the collection of data on the use of AI in a collaborative setting, including how AI affects communication and teamwork among team members, and how it impacts the quality and perceived creativity of the resulting comic.

In section 4.2.5 I identified that people not only form views about their own relationship with AI but also about other individuals relationship with AI. This suggests that the use of AI in creative work is not just a technical issue but is also tied to social dynamics and identity formation within the creative community. Therefore, further future studies could look into exploring how this ingroup and outgroup dynamic might influence the adoption and use of AI in the creative process.

Finally, the literature review conducted in Chapter 2 revealed that only a limited number of publicly available AI-tools are designed to assist users with specific tasks. Majority of well known AI tools that are readily available to the public, are either designed for research purposes [55, 61] or do not have a specific niche they cater towards. My project shows that there is a market for AI-tools that assist users in specific fields like creating comics. Potential tools could include an AI-tool specifically designed to help international students correct their grammar in their writing. It would be interesting to explore how this is any different or better than using existing tools such as Grammarly.

Chapter 7

Conclusion

7.1 Summary

The aim of this project was to examine the potential of AI in the comic creation process and assess its impact on enhance creativity and innovation. For achieve this, I conducted an extensive literature review to gain an understanding on people's perceptions of AI and on existing AI-based creativity support tools for writing and creating art. From this review, I determined that there was a lack of studies on the integration of AI could into the comic creation process and the how comic creators to such tools. Subsequently, I developed an AI-assisted comic creation tool, ComicPal. During the development, I iteratively sought feedback from users to improve the tool. Fifteen participants were recruited to test the first version of ComicPal, where they used the tool for 1-2 weeks to create a short comic story. Once done, they were asked to complete a questionnaire, which that was then analysed using both quantitative and qualitative analysis methods(thematic analysis). From the results, I found that ComicPal version 1 was generally positively received by users and was successful in enhancing their creativity. The analysis also indicated areas for improvement and new features that users requested, which were used to developed ComicPal version 2.

A focus group study was then conducted to examine the interaction dynamics between creators and AI during the comic creation process. It also shed light onto possible barriers to adopting AI as a creativity tool for making comics. Four individuals were recruited to use ComicPal to create a short comic, after which they engaged in a group discussion about their experiences via Microsoft Teams. The transcript was analysed using inductive coding and thematic analysis was performed using a software called Delve. The analysis revealed that creators' interaction dynamics with the AI can influenced by their motivations, values and even on the specific project they are working on. Participants who prioritised productivity in the moment were more willing to give up some autonomy to the AI, while those who valued the human-touch were more likely to grow frustrated when the AI failed to align with their artistic intent. Additionally, the AI's ability to help generate ideas and inspire creators was greatly appreciated and even the sometimes absurd outputs were appreciated by the user. however, although the participants were understanding of the benefits that AI could bring to comic creators, they all held some form of reservation that held them back from fully accepting AI as a creativity tool. These reservations revolved around concerns about copyright, fears of being replaced, and the idea that undeserving individuals could benefit from the technology at the expense of hardworking creators. These reservations were sometimes rooted in personal negative experiences or were anecdotal evidence of other creator's negative experiences with AI.

The results of this study demonstrate AI-CSTs can help alleviate certain challenges comic creators face. Therefore, these findings offer valuable insights for the design of future AI-CSTs for comic creation, with a focus on accommodating creators' changing needs and encouraging experimentation.

Bibliography

[1] ARTCRUMBS . Creator tips and tricks 17: Recovering from blocks and burnout, 08 2022. URL: https://globalcomix.com/news/details/342/creator-tips-and-tricks-17-recovering-from-blocks-and-burnout.

[2] Posted bybryantalpha . Finding the balance between words, pictures and time in making comics, 07 2010. URL: https://bryantpauljohnson.com/2010/07/08/finding-the-balance-between-words-pictures-and-time-in-making-comics/.

[3] aicomicbooks. aicomicbooks, 2022. URL: https://aicomicbooks.com/.

[4] Giana Angelo. Cognitive function in brief, 04 2017. URL: https://lpi.oregonstate.edu/mic/health-disease/cognitive-function-in-brief#:~:text=Cognition%20basically%20means%20using%20your.

[5] Theo Araujo, Natali Helberger, Sanne Kruikemeier, and Claes de Vreese. In ai we trust? perceptions about automated decision-making by artificial intelligence. *AI SOCIETY*, 35, 09 2020. doi:10.1007/s00146-019-00931-w.

[6] Jeffrey Bardzell. Creativity in amateur multimedia : Popular culture, critical theory, and hci. *Human Technology: An Interdisciplinary Journal on Humans in ICT Environments*, 3, 02 2007. doi:10.17011/ht/urn.200768.

[7] Oloff C. Biermann, Ning F. Ma, and Dongwook Yoon. From tool to companion: Storywriters want ai writers to respect their personal values and writing strategies. DIS '22, page 1209–1227, New York, NY, USA, 2022. Association for Computing Machinery. doi:10.1145/3532106.3533506.

[8] BLOOM. Bloom, 2020. URL: https://huggingface.co/bigscience/bloom.

[9] Boyd Branch, Piotr Mirowski, and Kory W. Mathewson. Collaborative storytelling with human actors and ai narrators, 2021. arXiv:2109.14728.

[10] Virginia Braun and Victoria Clarke. Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3:77–101, 01 2006. doi:10.1191/1478088706qp063oa.

[11] chatgpt. chatgpt, 2022. URL: https://openai.com/blog/chatgpt.

[12] John Joon Young Chung. Artistic user expressions in ai-powered creativity support tools. In *Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, UIST '22 Adjunct, New York, NY, USA, 2022. Association for Computing Machinery. doi:10.1145/3526114.3558531.

[13] John Joon Young Chung, Shiqing He, and Eytan Adar. The intersection of users, roles, interactions, and technologies in creativity support tools. In *Designing Interactive Systems Conference 2021*, DIS '21, page 1817–1833, New York, NY, USA, 2021. Association for Computing Machinery. doi:10.1145/3461778.3462050.

[14] John Joon Young Chung, Shiqing He, and Eytan Adar. Artist support networks: Implications for future creativity support tools. In *Designing Interactive Systems Conference*, DIS '22, page 232–246, New York, NY, USA, 2022. Association for Computing Machinery. doi:10.1145/3532106.3533505.

[15] John Joon Young Chung, Wooseok Kim, Kang Min Yoo, Hwaran Lee, Eytan Adar, and Minsuk Chang. Talebrush: Sketching stories with generative pretrained language models. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, CHI '22, New York, NY, USA, 2022. Association for Computing Machinery. doi:10.1145/3491102.3501819.

[16] Kate Compton and Michael Mateas. Casual creators. In *International Conference on Innovative Computing and Cloud Computing*, 2015.

[17] Robert Dale. Gpt-3: What's it good for? *Natural Language Engineering*, 27(1):113–118, 2021. doi:10.1017/S1351324920000601.

[18] Dalle. Dalle, 2021. URL: https://openai.com/product/dall-e-2.

[19] Antonio Daniele and Yi-Zhe Song. Ai + art = human. AIES '19, page 155–161, New York, NY, USA, 2019. Association for Computing Machinery. doi:10.1145/3306618.3314233.

[20] Nicholas Davis, Chih-Pin Hsiao, Yanna Popova, and Brian Magerko. *An Enactive Model of Creativity for Computational Collaboration and Co-creation*. 01 2015. doi:10.1007/978-1-4471-6681-8_7.

[21] Nicholas Davis, Chih-PIn Hsiao, Kunwar Yashraj Singh, Lisa Li, and Brian Magerko. Empirically studying participatory sense-making in abstract drawing with a co-creative cognitive agent. In *Proceedings of the 21st International Conference on Intelligent User Interfaces*, IUI '16, page 196–207, New York, NY, USA, 2016. Association for Computing Machinery. doi:10.1145/2856767.2856795.

[22] DeepAI. Deepai, 2017. URL: https://deepai.org/.

[23] Delve. Delve, 2020. URL: https://delvetool.com/blog/thematicanalysis.

[24] Daniel Dixon, Manoj Prasad, and Tracy Hammond. Icandraw? - using sketch recognition and corrective feedback to assist a user in drawing human faces. volume 2, pages 897–906, 04 2010. doi:10.1145/1753326.1753459.

[25] dndkit. dndkit, 2023. URL: https://docs.dndkit.com/.

[26] Charlotte Doyle. Creative flow as a unique cognitive process. *Frontiers in Psychology*, 8, 08 2017. doi:10.3389/fpsyg.2017.01348.

[27] Sylvester Ebigbagha. Overcoming creative block and generating innovative ideas for development: Heuristics for art/design. 8:1–024, 07 2019.

[28] Benj Edwards. Ai-generated comic artwork loses us copyright protection, 02 2023. URL: https://arstechnica.com/information-technology/2023/02/us-copyright-office-withdraws-copyright-for-ai-generated-comic-artwork/.

[29] Will Eisner. Comics and sequential art: Principles and practices from thelegendary cartoonist. 2008.

[30] Ziv Epstein, Sydney Levine, David G. Rand, and Iyad Rahwan. Who gets credit for ai-generated art? *iScience*, 23(9):101515, 2020. URL: https://www.sciencedirect.com/science/article/pii/S2589004220307070, doi:https://doi.org/10.1016/j.isci.2020.101515.

[31] hyperbolic films. I made a comic book using ai generated art in midjourney, 09 2022. URL: https://www.youtube.com/watch?v=BGNHq4aNUSo&t=329s.

[32] Herbert Freudenberger. Staff burn-out. *Journal of Social Issues*, 30:159 – 165, 04 2010. doi:10.1111/j.1540-4560.1974.tb00706.x.

[33] Maliheh Ghajargar, Jeffrey Bardzell, and Love Lagerkvist. A redhead walks into a bar: Experiences of writing fiction with artificial intelligence. pages 230–241, 11 2022. doi:10.1145/3569219.3569418.

[34] David Gianatasio. A creative director and an a.i. made this comic book together, 09 2022. URL: https://musebycl.io/digital-data/creative-director-and-ai-made-comic-book-together.

[35] Contact Graphixly. Creating flat colors for comics and illustrations, 05 2020. URL: https://graphixly.com/blogs/news/creating-flat-colors-for-comics-and-illustrations#:~:text=In%20comic%20books%20that%20are.

[36] grekoprinting comixwellspring. Create a comic: How to plan and lay out your comic, 08 2022. URL: https://www.grekoprinting-comixwellspring.com/blog/create-comic-how-plan-lay-out-your-comic/.

[37] Jeffrey Heer. Agency plus automation: Designing artificial intelligence into interactive systems. *Proceedings of the National Academy of Sciences*, 116:1844 – 1850, 2019.

[38] Austin Henderson. Interaction design. *Ubiquity*, 2002:2, 03 2002. doi:10.1145/512526.512528.

[39] html2canvas. html2canvas, 2023. URL: https://www.npmjs.com/package/html2canvas.

[40] Joy Kim, Avi Bagla, and Michael S. Bernstein. Designing creativity support tools for failure. In *Proceedings of the 2015 ACM SIGCHI Conference on Creativity and Cognition*, CC '15, page 157–160, New York, NY, USA, 2015. Association for Computing Machinery. doi:10.1145/2757226.2764542.

[41] Scott Kuindersma, Robin Deits, Maurice Fallon, Andrés Valenzuela, Hongkai Dai, Frank Permenter, Twan Koolen, Pat Marion, and Russ Tedrake. Optimization-based locomotion planning, estimation, and control design for the atlas humanoid robot. *Autonomous Robots*, 40, 07 2015. doi:10.1007/s10514-015-9479-3.

[42] Love Lagerkvist and Maliheh Ghajargar. Multiverse: Exploring human machine learning interaction through cybertextual generative literature. 10 2020. doi:10.1145/3423423.3423424.

[43] LaMDA. Lamda, 2021. URL: https://blog.google/technology/ai/lamda/.

[44] Devin Larson. Overview of the comic creation process, 01 2014. URL: https://www.makingcomics.com/2014/01/16/overview-comic-creation-process/.

[45] Yong Jae Lee, C. Zitnick, and Michael Cohen. Shadowdraw: Real-time user guidance for freehand drawing. *ACM Trans. Graph.*, 30:27, 07 2011. doi:10.1145/2010324.1964922.

[46] Shane Legg and Marcus Hutter. A collection of definitions of intelligence. *Advances in Artificial General Intelligence: Concepts, Architectures and Algorithms*, 157, 07 2007.

[47] Q. Vera Liao, Daniel Gruen, and Sarah Miller. Questioning the AI: Informing design practices for explainable AI user experiences. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, apr 2020. URL: https://doi.org/10.1145%2F3313831.3376590, doi:10.1145/3313831.3376590.

[48] Maria Teresa Llano, Mark d'Inverno, Matthew Yee-King, Jon McCormack, Alon Ilsar, Alison Pease, and Simon Colton. Explainable computational creativity, 2022. arXiv:2205.05682.

[49] Oxford Internet Institute logo. Oii — creativity support tools: A grand challenge, 06 2006. URL: https://www.oii.ox.ac.uk/news-events/videos/creativity-support-tools-a-grand-challenge/#:~:text=Creativity%20Support%20Tools%20is%20a.

[50] Brian Lubars and Chenhao Tan. Ask not what ai can do, but what ai should do: Towards a framework of task delegability, 2019. arXiv:1902.03245.

[51] Christina Maslach and Michael Leiter. Understanding the burnout experience: Recent research and its implications for psychiatry. *World Psychiatry*, 15:103–111, 06 2016. doi:10.1002/wps.20311.

[52] Scott Mccloud. *Reinventing comics : the evolution of an art form*. Perennial, 2000.

[53] Scott Mccloud. *Making comics : storytelling secrets of comics, manga and graphic novels*. Harper, 2006.

[54] Mckinsey. Global survey: The state of ai in 2021 — mckinsey, 2021. URL: https://www.mckinsey.com/capabilities/quantumblack/our-insights/global-survey-the-state-of-ai-in-2021.

[55] Midjourney. Midjourney, 2022. URL: https://www.midjourney.com/.

[56] Jim Clyde Monge. Midjourney ai's massive improvements with new v3 algorithm, 07 2022. URL: https://medium.com/mlearning-ai/midjourney-ais-massive-improvements-with-new-v3-algorithm-a44440fb21a8.

[57] Reem Nadeem. Public awareness of artificial intelligence in everyday activities, 02 2023. URL: https://www.pewresearch.org/science/2023/02/15/public-awareness-of-artificial-intelligence-in-everyday-activities/.

[58] Karim Nader, Paul Toprac, Suzanne Scott, and Samuel Baker. Public understanding of artificial intelligence through entertainment media. *AI SOCIETY*, 04 2022. doi:10.1007/s00146-022-01427-w.

[59] Changhoon Oh, Jungwoo Song, Jinhan Choi, Seonghyeon Kim, Sungwoo Lee, and Bongwon Suh. I lead, you help but only with enough details: Understanding user experience of co-creation with artificial intelligence. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, CHI '18, page 1–13, New York, NY, USA, 2018. Association for Computing Machinery. doi:10.1145/3173574.3174223.

[60] Lawrence J. Oliver. Helping students overcome writer's block. *Journal of Reading*, 26(2):162–168, 1982. URL: http://www.jstor.org/stable/40029248.

[61] OpenAI. Openai, 2015. URL: https://openai.com/.

[62] openaisdk. openaisdk, 2023. URL: https://www.npmjs.com/package/openai.

[63] Tomohiro Osaki. Japan's first ai manga has people asking: Is it machine magic or art menace?, 03 2023. URL: https://www.japantimes.co.jp/life/2023/03/07/digital/japans-first-ai-manga-people-asking-machine-magic-art-menace/.

[64] PerfectFreehand. Perfectfreehand, 2022. URL: https://www.npmjs.com/package/perfect-freehand.

[65] Daniel Pinto dos Santos, Daniel Giese, S. Brodehl, Seung-Hun Chon, W. Staab, Robert Kleinert, D. Maintz, and Bettina Baeßler. Medical students' attitude towards artificial intelligence: a multicentre survey. *European Radiology*, 29, 07 2018. doi:10.1007/s00330-018-5601-1.

[66] reacthookforms. reacthookforms, 2023. URL: https://www.npmjs.com/package/react-hook-form.

[67] Melissa Roemmele and Andrew Gordon. Creative help: A story writing assistant. pages 81–92, 11 2015. doi:10.1007/978-3-319-27036-4_8.

[68] Kevin Roose. An a.i.-generated picture won an art prize. artists aren't happy. *The New York Times*, 09 2022. URL: https://www.nytimes.com/2022/09/02/technology/ai-artificial-intelligence-artists.html.

[69] roughjs. roughjs, 2020. URL: https://roughjs.com/.

[70] Quentin Roy, Daniel Vogel, and Futian Zhang. Automation accuracy is good, but high controllability may be better. 03 2019. doi:10.31219/osf.io/ajh9n.

[71] Ben Shneiderman. Creativity support tools: Accelerating discovery and innovation. *Commun. ACM*, 50(12):20–32, dec 2007. doi:10.1145/1323688.1323689.

[72] Cornelia Sindermann, Yang Haibo, Jon Elhai, Shixin Yang, Ling Quan, Mei Li, and Christian Montag. Acceptance and fear of artificial intelligence: associations with personality in a german and a chinese sample. *Discover Psychology*, 2, 12 2021. doi:10.1007/s44202-022-00020-y.

[73] Omar Spahi. How to break into comics: Networking, 10 2018. URL: https://www.comicon.com/2018/10/19/how-to-break-into-comics-networking/.

[74] StableDiffusion. Stablediffusion, 2022. URL: https://stablediffusionweb.com/.

[75] Michael Cheng tek Tai. The impact of artificial intelligence on human society and bioethics. *Tzu-Chi Medical Journal*, 32:339 – 343, 2020.

[76] tiptap. tiptap, 2023. URL: https://tiptap.dev/.

[77] Ma. Dolores C. Tongco. Purposive sampling as a tool for informant selection. *Ethnobotany Research and Applications*, 5:147–158, 2007.

[78] vercel. vercel, 2015. URL: https://vercel.com/.

[79] Jamie Ward, Daisy Thompson-Lake, Roxanne Ely, and Flora Kaminski. Synaesthesia, creativity and art: What is the link? *British journal of psychology (London, England : 1953)*, 99:127–41, 03 2008. doi:10.1348/000712607X204164.

[80] Webcomics Digest WebcomicsFeed. Where to publish your webcomics, 06 2021. URL: https://webcomicsfeed.medium.com/where-to-publish-your-webcomics-c2d6067898b2#:~:text=Social%20Media%20Platforms&text=For%20social%20media%20Instagram%20is.

[81] Geraint Wiggins. A preliminary framework for description, analysis and comparison of creative systems. *Knowledge-Based Systems*, 19:449–458, 11 2006. doi:10.1016/j.knosys.2006.04.009.

[82] Chuan Yan, John Joon Young Chung, Yoon Kiheon, Yotam Gingold, Eytan Adar, and Sungsoo Ray Hong. Flatmagic: Improving flat colorization through ai-driven design for digital comic professionals. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, CHI '22, New York, NY, USA, 2022. Association for Computing Machinery. doi:10.1145/3491102.3502075.

Appendix A

ComicPal

A.1 Webapp

https://comicpal.vercel.app/

A.2 Video

https://youtu.be/Dqs-RYhIk44

A.3 Source Code

https://github.com/KaliKaloo/ComicPal

Appendix B

Participation Sheet

University of BRISTOL

Participant Information Sheet

Project title: Exploring the Potential of Artificial Intelligence in Empowering Comic Creators: Enhancing the Creative Process and Promoting Innovation

Invitation paragraph
I would like to invite you to take part in my final year research project aiming to investigate how AI can be used to enhance the creative processes of comic book creators. Before you decide whether or not to participate, I would like you to understand why the research is being conducted and what it would involve for you. Talk to others about the study if you wish. Please ask me questions if anything is unclear.

What is the purpose of the project?
In recent times, there has been a growing familiarity with the concept of artificial intelligence (AI) among the general public. Although a universally accepted definition for AI remains elusive, it is typically described as "the theory and development of computer systems that possess the ability to perform tasks that would normally require human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages". Several AI systems, including DALE, Midjourney, and ChatGPT, have become household names, reflecting the widespread reach and impact of this technology. However, due to the general public's lack of comprehension of how AI works, there is a growing fear that AI will soon displace humans in various job sectors. This apprehension is particularly pronounced in the art community, where many artists have publicly expressed their opposition to the use of AI in creative endeavours. This resistance is not unfounded, as some nefarious actors have exploited AI to deceive and steal others' works. Regrettably, this has resulted in a taboo surrounding any discussion of AI, preventing a productive dialogue on the subject.

The objective of this study is to explore the present and potential impact of AI on creative endeavours, with a particular emphasis on comic art and writing. The study aims to establish the essential design principles for a productive creative-AI-tool that facilitates human creativity.

Why have I been invited to participate?
We are reaching out to you because we value your contributions to the field of art and/or writing. We would be grateful to converse with you and gain insights on how artificial intelligence could enhance your creative process.

Do I have to take part?
No, your participation is entirely voluntary. We will describe the study and go through this information sheet with you before you participate, and answer any questions you might have. If you agree to take part, we will then ask you to sign a consent form. You are free to withdraw at any time, without giving a reason. If you wish to retract your data, the deadline for this will be one week after data collection.

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Figure B.1: Participation Sheet Page 1

What will happen to me if I take part and what will I have to do?
This project will collect data though semi-structured interviews which will last about an hour and guided by the researcher. These will take place online through MS-Teams or in-person interviews conducted in private rooms at University of Bristol. Only a digital voice recorder will be used to capture a transcript of the meeting for the purpose of extracting data afterwards. Prior to the meeting, you will be asked to test out a website for around a week. During the interview, initial questions will go over your current creative workflow, then transition into your experiences using the website.

What are the possible disadvantages and risks involved in taking part in the project?
We do not anticipate any physical or legal risk.

What are the possible benefits of taking part?
Your first-hand experiences as a creative can enrich our understanding of the use of AI in the comic book industry. Your insights could shed light on how collaborations between humans and AI can enhance the creative process.

Will my participation in this project be kept confidential?
Yes. All of your data will be kept anonymised and identifiable information will be removed within two weeks of collection. The research data will only be used for research purpose. All data will be stored securely and in accordance with data protection regulations of the University of Bristol and Data Protection Act (GDPR). It will not be shared with anyone other than the researcher and the research supervisors.

What will happen to the results of the research project?
The result will be published as a final year dissertation at the University of Bristol which will be made available to participants. You will not be identified as a named individual in any report/publication.

Who is organising and funding the research?
The research is being done as part of a Computer Science (MEng) Undergraduate Dissertation at the University of Bristol.

Who has reviewed the study?
This research has been reviewed by supervisors Peter Bennett and Sarah L Connolly who are both part of the Department of Computer Science at University of Bristol.

Further information and contact details
Participants may want further information in relation to the study and may have questions about what they are being asked to do, so the researcher's contact details should be included here.

If you have any questions about the project, please contact Pragya Gurung (researcher) wq19451@bristol.ac.uk or Peter Bennet (lead project supervisor) Peter.Bennett@bristol.ac.uk

If participants have any concerns related to their participation in this study contact the Faculty of Engineering Research Ethics Committee, via Liam McKervey, Research Governance and Ethics Officer (Tel: 0117 331 7472 email: Liam.McKervey@bristol.ac.uk)

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Figure B.2: Participation Sheet Page 2